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1. Introduction

1.1 Background

Remittance: A remittance is a non-commercial transfer of money by a foreign worker, a member of a diaspora community, or a citizen with familial ties abroad, for household income in their home country or homeland.

Money sent home by migrants competes with international aid as one of the largest financial inflows to developing countries. Remittance is more than three times as large as the total global foreign aid. In 2021, $780 billion was sent to 800 million people, while foreign aid totalled $200 billion. Most remittance flows from high-income countries to lower-income countries.Workers' remittances are a significant part of international capital flows, especially with regard to labor-exporting countries.Remittance has been defined by the World Bank as the part of the earnings which a migrant worker sends back to family members in the country of origin. Worldwide, the flow of remittance has increased from US$72.3 billion in 2001 to approximately US$483 billion in 2011. According to the World Bank, in 2018 overall global remittance grew 10% to US$689 billion, including US$528 billion in 2019 to developing countries. Overall global remittance was expected to grow 3.7% to US$715 billion in 2019, including US$549 billion to developing nations.According to the World Bank, remittance flows to low- and middle-income countries grew by about five percent in 2022, reaching US$626 billion.

GDP:Gross domestic product (GDP) is a monetary measure of the total market value of all of the final goods and services which are produced and rendered during a specific period of time (usually a year) by a country or countries. GDP is often used to measure the economic activity of a country or region. The major components of GDP are consumption, government spending, net exports (exports minus imports), and investment. Changing any of these factors can increase the size of the economy. For example, population growth through mass immigration can raise consumption and demand for public services, thereby contributing to GDP growth. However, GDP is not a measure of overall standard of living or well-being, as it does not account for how income is distributed among the population.

Inflation:In economics, **inflation** is an increase in the average price of goods and services in terms of money. This increase is measured using a price index, typically a consumer price index (CPI). When the general price level rises, each unit of currency buys fewer goods and services; consequently, inflation corresponds to a reduction in the purchasing power of money.The common measure of inflation is the **inflation rate**, the annualized percentage change in a general price index.

Changes in inflation are widely attributed to increases in the money supply, fluctuations in real demand for goods and services (a1lso known as demand shocks, including changes in fiscal or monetary policy), changes in available supplies such as during energy crises (also known as supply shocks), significant decreases in interest rates set by the central bank, or changes in inflation expectations, which may be self-fulfilling.

Exchange rates:In finance, an **exchange rate** is the rate at which one currency will be exchanged for another currency. Currencies are most commonly national currencies, but

may be sub-national as in the case of Hong Kong or supra-national as in the case of the euro.Each country determines the exchange rate regime that will apply to its currency. For example, a currency may be floating, pegged (fixed), or a hybrid. Governments can impose certain limits and controls on exchange rates. Countries can also have a strong or weak currency. There is no agreement in the economic literature on the optimal national exchange rate policy (unlike on the subject of trade where free trade is considered optimal). Rather, national exchange rate regimes reflect political considerations.In floating exchange rate regimes, exchange rates are determined in the foreign exchange market, which is open to a wide range of different types of buyers and sellers, and where currency trading is continuous: 24 hours a day except weekends (i.e. trading from 20:15 GMT on Sunday until 22:00 GMT Friday). The spot exchange rate is the current exchange rate, while the forward exchange rate is an exchange rate that is quoted and traded today but for delivery and payment on a specific future date.

Net migration:The **net migration rate** is the difference between the number of immigrants (people coming into an area) and the number of emigrants (people leaving an area) per year divided by the population. When the number of immigrants is larger than the number of emigrants, a positive net migration rate occurs. A positive net migration rate indicates that there are more people entering than leaving an area. When more emigrate from a country, the result is a negative net migration rate, meaning that more people are leaving than entering the area. When there is an equal number of immigrants and emigrants, the net migration rate is balanced.

The net migration rate is calculated over a one-year period using the mid year population and a ratio.
Over the past few years, remittance inflow in Nepal has shown continuous growth. It contributes to national income, supports household expenses, education, healthcare, and investments, and influences broader economic indicators such as Gross Domestic Product (GDP), inflation, and employment. However, dependency on remittance may also create challenges, making it important to understand its overall impact on Nepal's economy.

This project aims to analyze the relationship between remittance and key economic indicators in Nepal. The study will use historical data to examine how remittance affects GDP growth, inflation, and other economic variables. Understanding these relationships can provide useful insights for economic planning and policy development.

1.2 Statement of the Problem

While the microeconomic benefits of remittance including but not limited to immediate poverty reduction, smoothing household consumption, as well as improving human capital access are clear, their structural macroeconomic impact presents a complex policy puzzle.

On one hand, remittance supplies critical foreign exchange reserves, which helps to finance a widening trade deficit and support the national economy. On the other hand, an economy that is heavily reliant on external inflows is very susceptible to sudden global market shocks, changes in destination-country immigration laws, and regional economic downturns. Furthermore, sustained, massive capital injections risk driving Dutch Disease effects: inflating the price of non-tradable goods (like real estate and domestic services),

shifting labor away from productive sectors like agriculture, and creating a consumption-driven economy highly dependent on imports rather than domestic manufacturing.

1.3 Literature Review

1.3.1 Theoretical Underpinnings of Remittances

In macroeconomic theory, remittance inflows are typically analyzed through two distinct lenses:

- The Optimistic Growth Model: This framework treats remittances as a stable capital injection that helps overcome domestic credit constraints. In developing countries with shallow financial markets, these inflows serve as an informal source of investment capital, funding small-scale businesses, agricultural upgrades, and human capital development (education and healthcare), which ultimately expands aggregate supply.
- The Pessimistic Structuralist Model: This model views remittances as a driver of consumption dependency. It argues that the vast majority of incoming transfers are spent on non-productive, imported consumer goods or real estate speculation. Consequently, they fail to build domestic industrial capacity, can fuel local demand-pull inflation, and may lower domestic labor force participation by acting as an unearned income cushion.

1.3.2 The Dutch Disease Paradigm

A key theoretical foundation for this study is the Dutch Disease model developed by Corden and Neary (1982). When a country experiences a sustained surge in foreign currency inflows (like a remittance boom), the economy experiences two core adjustments:

- The Spending Effect: The large injection of foreign funds boosts aggregate demand. Because the prices of tradable goods (imports/exports) are anchored by international markets, this extra demand bids up the prices of non-tradable goods (housing, services), causing the real exchange rate to appreciate.
- The Resource Movement Effect: Rising profitability and wages in the non-tradable sector draw mobile factors of production (such as labor and capital) away from traditional tradable sectors like domestic manufacturing and agriculture, potentially weakening long-term structural competitiveness.

1.4 Project Scope and Application

This project has many far reaching applications in the macroeconomic and policy making landscape of remittance reliant nations like Nepal. It can be used directly by the central

banking authorities and fiscal planners of a nation to predict systemic liquidity change and optimize national reserves. The analytical framework developed here is limited to a particular study of formal transfer mechanisms, but it can be extended to analyse transmission mechanisms of external shocks in a wider setting for small open economies. It provides an empirical basis for time series modelling to researchers and academic bodies with validated baseline parameters, unit root integrations and lag structures to test more complex non-linear models or structural breaks.

The project provides a structural mapping of the complex relationship between remittance inflows and key macroeconomic variables, highlighting important dynamics including the strong multiplier effect on national output. The project's demonstrated nominal coefficient of 2.34x in output expansion provides a data-driven case for government agencies to design targeted financial instruments such as diaspora bonds or municipal infrastructure pools. equips national development planners with the ability to design policies that redirect financial capital from speculative real estate and imported consumer goods to productive national infrastructure, such as transport networks and sustainable energy systems. Furthermore, the project enables monetary authorities to quantify the structural elasticity between exchange rate fluctuations and incoming liquidity, thereby providing them with the necessary insights to maintain essential currency pegs and develop countercyclical tax regimes to mitigate trade deficits.

The project extensively deals with the macroeconomic scenario of Nepal considering it as an open economy with heavy dependence on external labour export channels. However, this analysis is restricted to a 32-year annual time series from 1993 to 2025 with the dataset capturing structural shifts from post-1990 economic liberalisation to the expansion of labour migration to the Gulf Cooperation Council (GCC) countries and Malaysia. The scope is limited strictly to the quantitative analysis of 5 core variables: Total nominal Remittance Inflows, GDP, Nominal Exchange Rates (USD to NPR), Net Outbound Migration counts and Consumer Price Inflation. It thus does not reflect informal financial networks like Hundi due to limitations in data tracking, nor does it capture microeconomic dimensions like household level survey data, behavioural consumption choices or regional patterns of wealth distribution.

2. Objectives

The main objectives of this project are:

- To analyze the trend of remittance inflow in Nepal over time
- To study the relationship between remittance and Gross Domestic Product (GDP) in Nepal
- To examine the relationship between remittance and inflation
- To study the relationship between migration rate and remittance
- To use statistical techniques to forecast remittance inflow trends
- To understand the economic importance of remittance inflow in Nepal

3. Data

The data we created using various data sources

Table 1:Dataset

Year	Remittance	GDP	Net migration	Inflation	Exchange rate(US$ to NPR)
1993	54825103.8	3660041667	-37782	7.505394	62.75
1994	50118518.8	4066775510	-27415	8.349287	62.75
1995	56822666.9	4401104418	-86232	7.6229696	62.75
1996	44160126.1	4521580381	-92307	9.220467	62.75
1997	49458057.9	4918691917	-98454	4.009989	62.75
1998	67504902.9	4856255044	-108400	11.244468	66.32
1999	83462780.3	5033642384	-122551	7.4511127	68.05
2000	111498192	5494252208	-147193	2.4788203	71.49
2001	146985152	6007055042	-148973	2.6883037	75.47
2002	678485892	6050875807	-153730	3.0293994	78.29
2003	771072126	6330473097	-168314	5.7070093	76.26
2004	822608955	7273938315	-197336	2.8418114	73.93
2005	1211823253	8130258378	-211024	6.836333	71.72
2006	1453231426	9043715356	-241455	6.920336	73.12
2007	1733858518	10325618017	-252526	2.2692192	66.69
2008	2727140487	12545438605	-261540	9.90783	70.01
2009	2983336877	12854985464	-266144	11.094824	77.78
2010	3464092682	16002656434	-269963	9.326504	73.46
2011	4216894010	21573872421	-347552	9.227076	74.9
2012	4792506200	21703100877	-371707	9.45981	85.73
2013	5583702128	22162204925	-373064	9.040163	93.93
2014	5888650155	22731612922	-310156	8.364155	97.92
2015	6729935671	24360801287	-221062	7.868909	102.92
2016	6611838544	24524109484	-341748	8.790343	107.78
2017	6928133767	28971588940	-368787	2.77704	104.5
2018	8286629516	33111525237	-295151	4.4065943	109.71
2019	8244465869	34186180699	182203	5.56616	112.94
2020	8107744662	33433659301	162139	5.0563583	118.87
2021	8225960575	36924841394	130001	4.1267676	118.57
2022	9411659366	41182939601	-379321	7.6702056	126.04
2023	1.0764E+10	41047772331	-409782	7.1158724	132.44
2024	1.1254E+10	42914268287	-401282	4.6850967	134.17

2025	1.8911E+11	**45,84,40,00,000**	-3,64,699	4.06	139.37

3.1. Data Collection and Sources

This study relies entirely on secondary time-series data to fulfill the research objectives. To ensure a high degree of validity, accuracy, and reliability, data have been compiled from reputable national and international macroeconomic databases. The dataset consists of yearly records tracking remittance inflows alongside primary socio-economic indicators in the context of Nepal.

The specific variables, their definitions, and their corresponding official data repositories are outlined below:

1. Macroeconomic and Household Flow Indicators

- **Remittance Inflow:** Remittance inflow is measured as personal remittances received. This captures the total value of current transfers in cash or in kind made by nonresident households to resident households.
    - *Source:* World Bank Data360 (KNOMAD Database)
    - *URL:* https://data360.worldbank.org/en/indicator/WB_KNOMAD_MRI
- **Gross Domestic Product (GDP):** It is measured in current USD to quantify the overall economic output of Nepal and to assess the ratio and impact of remittance relative to national economic expansion.
    - *Source:* World Bank DataBank (World Development Indicators)
    - *URL:* https://databank.worldbank.org/indicator/NY.GDP.MKTP.CD

2. Monetary and Price Stability Indicators

- **Inflation Rate:** Inflation rate is calculated via the Consumer Price Index (CPI) to gauge fluctuations in market purchasing power and price levels, which helps analyze the potential demand-pull inflation impacts of remittance.
    - *Source:* World Bank Data360 / International Monetary Fund (IMF WEO)
    - *URL:* https://data360.worldbank.org/en/indicator/IMF_WEO_PCPI
- **Exchange Rate:** Exchange rate is evaluated using the official foreign exchange rates (specifically USD to NPR) to understand how domestic currency valuation alters the real-value flow of incoming foreign remittances.
    - *Source:* Nepal Rastra Bank (NRB) Forex Portal
    - *URL:* https://www.nrb.org.np/forex

3. Demographic and Labor Indicators

- **Net Migration Rate:** Net migration rate tracks the net total of migrants during given periods (immigrants minus emigrants). This serves as a vital demographic proxy to determine how shifts in foreign employment numbers drive fluctuations in capital repatriation.

- *Source:* World Bank Open Data (UN Population Division Estimates)
- *URL:* https://data.worldbank.org/indicator/SM.POP.NETM?locations=NP

Summary of Variables and Sourcing Matrix

To streamline data tracking for statistical and econometric analysis, the dataset architecture is summarized in the table below:

Table 2:Summary of Variables

Variable Name	**Theoretical Unit**	**Primary Source**	**Access Node / Node URL**
Remittance Inflow	Current USD / % of GDP	World Bank (KNOMAD)	data360.worldbank.org
Gross Domestic Product	Current USD	World Bank (WDI)	databank.worldbank.org
Inflation Rate	Annual % (CPI)	IMF / World Bank	data360.worldbank.org
Exchange Rate	NPR per USD	Nepal Rastra Bank	nrb.org.np/forex
Net Migration	Total Count of Persons	World Bank (UNPD)	data.worldbank.org

4.METHODOLOGY

4.1 Research Methodology

This study adopts an empirical, **quantitative research design** operating within a **deductive framework**. The primary objective is to investigate the structural and macroeconomic relationships between remittance inflows, inflation, exchange rates, and net migration, with a specific focus on the economy of Nepal. Given that macro-level economic phenomena are best captured through quantifiable metrics, a quantitative methodology eliminates subjective bias and provides generalizable, reproducible insights.

The research is theoretically grounded in international trade and labor economics paradigms—specifically contrasting the *Optimistic Growth Model* against the *Pessimistic Structuralist Model* and the *Dutch Disease Paradigm*. By analyzing historical time-series data, this methodology seeks to determine whether empirical evidence supports the theory that massive resource injections (remittances) influence domestic structural competitiveness and migration dynamics.

4.2 Research Methods

4.2.1 Data Sources and Variables

To ensure academic validity and high reliability, this study utilizes secondary time-series data compiled from leading international organizations and national regulatory banks:

- **Remittance Inflows (X_1):** Extracted from the World Bank Data360 portal via the Global Knowledge Partnership on Migration and Development (KNOMAD) indicators (Indicator: WB_KNOMAD_MRI), measured in USD.

  World Bank

- **Inflation Rate (X_2):** Collected from the International Monetary Fund (IMF) World Economic Outlook (WEO) database (Indicator: IMF_WEO_PCPI), measuring the annual percentage change in the consumer price index.
- **Exchange Rate (X_3):** Retrieved directly from the official Forex historical database of the **Nepal Rastra Bank (NRB)**, representing the local currency (NPR) equivalent per unit of foreign currency.
- **Gross Domestic Product (X_4):** Sourced from the World Bank National Accounts Databank (Indicator: NY.GDP.MKTP.CD), representing the absolute economic output in current USD.
- **Net Migration (Y):** Sourced from demographic monitoring data to track the net flow of population movement (total immigrants minus total emigrants) over corresponding time intervals.

4.2.2 Data Cleaning and Pre-Processing

Before running statistical models or generating visual assets, the raw data underwent a structured data cleaning protocol:

1. **Format Unification:** Disparate source formats (.csv, .xlsx, and raw text queries) were standardized and compiled into a uniform master data matrix.
2. **Temporal Alignment:** Observations were synchronized using a single temporal index tracking fiscal/calendar years to form a cohesive time-series dataset.
3. **Handling Missing Data:** Null values and discrepancies at the temporal boundaries were audited. Minor gaps were resolved using linear interpolation where appropriate, while years with systemic missing entries were truncated to maintain statistical accuracy.
4. **Scaling Transformation:** Due to the large absolute values of variables like GDP and Remittances, the numbers were kept in their original scale but visualized using standard scientific notation (e.g., 1.5e+11) to preserve accuracy while eliminating graphic distortion.

4.2.3 Model Specification and Visualization

The foundational relationship highlighted in this stage of the project is evaluated using bivariate visual analysis and ordinary least squares (OLS) linear trend estimation:

$$\text{Net Migration}_t = \beta_0 + \beta_1(\text{Remittances}_t) + \varepsilon_t$$

Where:

- $\beta_0$represents the intercept.
- $\beta_1$represents the coefficient capturing the direction and magnitude of the effect of remittances on net migration.
- $\varepsilon_t$represents the stochastic error term at time t.

Analytical Software Environment

4.3. Model Specifications and Estimation Framework

4.3.1 Ordinary Least Squares (OLS) Regression Specifications

Four linear econometric models are specified to test specific macroeconomic hypotheses using stationary, differenced series to eliminate trend-driven overfitting:

4.3.1.1. Model 1 (Remittance Determinants Framework):

$$d_Remittance = \beta0 + \beta1(d_Net_mig) + \beta2(d_Ex_rate) + \varepsilon$$

This model estimates the immediate short-run impact of shifts in net migration and exchange rate movements on the change in formal remittance inflows.

4.3.1.2. Model 2 (Macroeconomic Growth Framework):

$$g_GDP = \beta0 + \beta1(g_Remittance) + \beta2(d_Ex_rate) + \varepsilon$$

This specification calculates the short-run economic growth elasticity with respect to remittance growth while controlling for contemporaneous exchange rate adjustments.

4.3.1.3. Model 3 (Inflationary Transmission Framework):

$$d_Inflation = \beta0 + \beta1(d_Remittance) + \varepsilon$$

This model tests for demand-side or cost-push inflationary pressures by regressing changes in the inflation rate directly on changes in remittance inflows.

4.3.1.4. Model 4 (Unconditional Growth Elasticity Framework):

$$g_GDP = \beta0 + \beta1(g_Remittance) + \varepsilon$$

This bivariate baseline isolates the direct, unconditional elasticity between remittance growth and national economic growth.

4.3.2 Multivariate Vector Autoregression (VAR) Framework

To evaluate dynamic, bidirectional dependencies and endogenous feedback loops within the system without imposing strict structural restrictions, an Optimized Vector Autoregression (VAR) framework is defined as follows:

$$Y_t = C + A_1(Y_\{t-1\}) + ... + A_p(Y_\{t-p\}) + u_t$$

Where Y_t is a vector containing the stationary endogenous variables [g_GDP, g_Remittance, d_Ex_rate]. To prevent underfitting (omitted-lag bias) or over-parameterization (overfitting), the optimal lag order ($p = 1$) is established mathematically via the VARselect() algorithm using Akaike and Bayesian Information Criteria (AIC/BIC).

4.3.3 Machine Learning and Univariate Forecasting Architectures

To cross-validate linear configurations against non-linear structures, two out-of-sample machine learning models are trained over an 80/20 train-test partition of the stationary dataset:

4.3.1. Decision Tree (rpart): To prevent algorithmic overfitting (training data memorization), the model is bound to a 5-fold cross-validation routine to isolate the optimal complexity parameter (cp).

4.3.2. Random Forest Ensemble: Built with 500 decision nodes to evaluate aggregate predictive accuracy and prevent localized variance overfitting.

For univariate time-series forecasting, an automated Autoregressive Integrated Moving Average (ARIMA) model is specified. The optimal parameters are selected by minimizing the corrected Akaike Information Criterion (AICc) to limit unnecessary parameter inflation.

4.4 System Configuration and Software Libraries

The quantitative analysis was developed within the R (Version 4.4.2) and RStudio environment, utilizing the following individual software libraries to execute the empirical pipeline:

- dplyr: Used to clean, transform, and restructure raw data vectors into stationary differences.
- readxl: Used to securely import the raw macroeconomic data from Excel worksheets into R.
- moments: Used to evaluate the structural distribution shapes of variables using skewness and kurtosis.
- ggplot2: Used to build professional, publication-ready line plots and diagnostic trend charts.
- corrplot: Used to map out directional patterns and visualize the initial correlation matrix.
- GGally: Used to generate advanced pairwise data grids to scan for structural patterns.
- stargazer: Used to format raw OLS outputs into clean, standardized academic regression tables.
- lmtest: Used to verify residual health via Breusch-Pagan and Durbin-Watson diagnostic tests.
- car: Used to calculate Variance Inflation Factors (VIF) to monitor multicollinearity.
- tseries: Used to execute Augmented Dickey-Fuller (ADF) tests to confirm time-series stationarity.

- caret: Used to partition data structures and manage machine learning control frameworks.
- randomForest: Used to construct the computational machine learning ensemble model.
- forecast: Used to model and predict future trends using historical macroeconomic trajectories.
- vars: Used to set up interdependent systems using Vector Autoregression frameworks.

5. Result and discussion:

Histogram is a plot used for distribution check in this project is used to visualize the empirical distribution and shape of a single variable. In histogram a symmetric bell curve denotes normality (Ideal for classical OLS assumptions), a histogram with long tail to the right represents a positive skewness (May require log-transform) and a histogram with a long tail to the left denotes negative skewness.

Time series line plot used in trend analysis is used to observe structural trends, cycles, and growth patterns over time. An Upward or Downward Trend in the plot indicates a strong signal of non stationarity or stochastic trend, a sudden sharp jumps or drops usually indicates a potential structural breaks (e.g., policy shifts, shocks, etc) and a constant mean and variance indicates visual indicator of a stationary series.

A scatter plot with linear fit was used for visual diagnostic. It is used to assess the direction, strength, and linearity between two variables. When the points in scatter plot are tightly clustered around an upward line indicates a strong positive linear relationship between the variables. When points are tightly clustered around a downward line, it indicates a strong negative linear relationship. When the scatter plot is wide and has random scattering, it indicates a weak or non-linear correlation between two variables and a shaded band around the line indicates 95% Confidence Interval for the predicted mean.

The correlation matrix plot was used for visual diagnosis in this project. It is used to diagnose pairwise relationships and screens for potential multi-collinearity. In correlation matrix darker, larger blue circles indicates a strong positive linear correlation, it occurs when r approaches +1. Darker, larger red circles points to a strong negative linear correlation, this usually occurs when r approaches -1 .A Pairwise correlations above |0.7| or |0.8 Independent variables suggest severe multicollinearity risk and require Variance Inflation Factor (VIF) verification.

Box and whisker plot was used for visual diagnosis in this project.. It is used to identify data dispersion, skewness, and statistical outliers. In the box and whisker plot if the center line is in median i.e. 50th percentile of the variable, box bounds represent Interquartile Range (IQR from 25th to 75th percentile) and whiskers indicate data reach within 1.5 times the IQR. Any isolated points beyond whiskers is a statistical outlier that might heavily distort OLS estimation lines.

OLS residual diagnostic i.e. Four-Panel Plot was used for visual diagnostic in this project.. It is used to: Tests core Gauss-Markov assumptions via linear model residuals. A plot with Residuals vs Fitted helps to test Linearity and homoscedasticity and it should look like a random, flat bounce. A "megaphone" shape implies Heteroskedasticity and a normal Q-Q is used to tests normality of errors. Dots should follow the straight diagonal line perfectly. S-shapes or curves imply non-normal residuals. Scale-Location is an alternative check for Heteroskedasticity, for it the line should be flat. And Residuals vs Leverage is used to Identify highly influential points. Points crossing outside the dashed red borders (Cook's Distance > 0.5 or 1.0) pull the line too much

Orthogonalized Impulse Response Function (IRF) was used for visual diagnostic in this project. It is used to trace the dynamic response of a target system variable over a time

horizon following a one-standard-deviation shock (innovation) in a specific variable. A solid line is expected dynamic path of the response variable, a dashed lines indicates 95% bootstrap confidence intervals. If the horizontal zero line lies completely outside the dashed lines, the dynamic impact is statistically significant

5.1 Graphs from Code

Boxplot of Remittance Inflows

Figure 1 presents a boxplot of raw remittance inflows to evaluate the data distribution and identify outliers. The highly compressed interquartile range (IQR) near the bottom reflects the low, stable baseline levels of formal remittances during the earlier years of the timeline. Crucially, a distinct, extreme upper outlier is visible near the 1.8×10^{11} mark, capturing the massive, exponential surge in recent volumes. This severe right-skewness confirms a non-normal distribution, validating the use of log-transformations and first-differencing in our econometric models to stabilize variance.

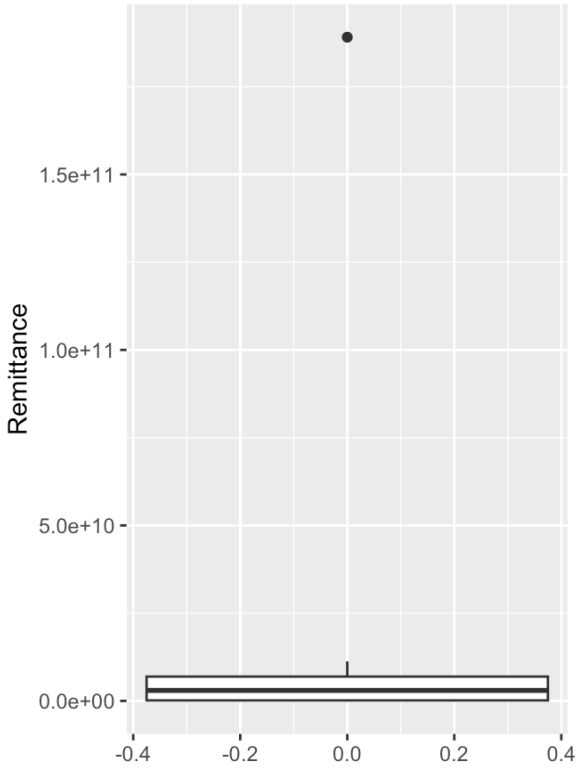


Figure 1: Box plot of remittance

Boxplot of Gross Domestic Product (GDP)

Figure 2 illustrates a boxplot of Nepal's raw Gross Domestic Product (GDP) to examine its statistical distribution and dispersion. Unlike the remittance series, the GDP data features a much broader interquartile range (IQR), highlighting steady, continuous structural growth over the sample period. The median line sits visibly closer to the lower quartile, and an elongated upper whisker extends up toward the 4.3×10^{10} mark. This pattern demonstrates a right-skewed, non-normal distribution driven by rapid economic expansion in the latter half of the timeline. The lack of isolated outlier points suggests a continuous macroeconomic trajectory, while the overall skewness confirms the need for data transformation (such as logging or differencing) to ensure stationarity in the subsequent econometric analysis.

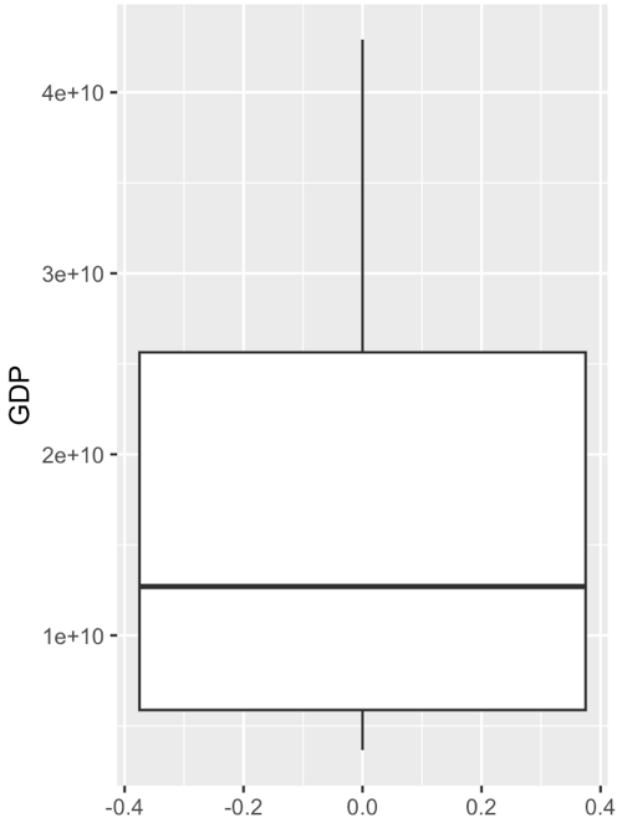


Figure 2: Box plot of GDP

Histogram and Frequency Distribution of Inflation Rate

Figure 3 illustrates a histogram detailing the empirical frequency distribution of Nepal's inflation rate over the sample period. The distribution spans from a lower bound near 2% up to a maximum interval around 12%, with a noticeable concentration of data points falling between the 7% and 10% blocks. The multi-modal shape of the histogram suggests that inflation has experienced distinct structural phases, likely corresponding to varying periods of domestic price shocks, global commodity price fluctuations, and peg-driven monetary transitions linked with the Indian Rupee. This dispersion highlights that while inflation remains volatile, it generally operates within a moderate-to-high single-digit baseline, confirming the necessity of evaluating its structural co-movements within the macro-econometric framework.

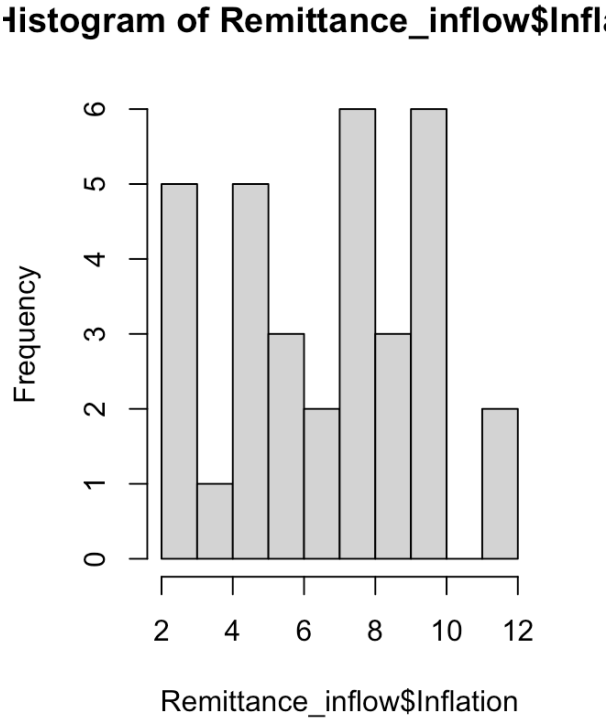


Figure 3: Histogram of Inflation

Histogram and Frequency Distribution of Remittance Inflows

Figure 4 illustrates a histogram tracking the empirical frequency distribution of raw remittance inflows. The data displays a severe positive (right) skew, with the vast majority of historical observations densely concentrated in the lowest bin near $0.0e+00$. Conversely, an isolated, solitary frequency bar sits at the extreme right of the spectrum, approaching the $2.0e+11$ mark. This stark structural polarization indicates that for most of the sample timeline, formal remittance inflows into Nepal remained at a low baseline, followed by an aggressive, exponential volume spike in recent years. The extreme non-normality and variance expansion captured here provide a robust empirical justification for applying logarithmic transformations or first-differencing to stabilize variance before running time-series econometric regressions.

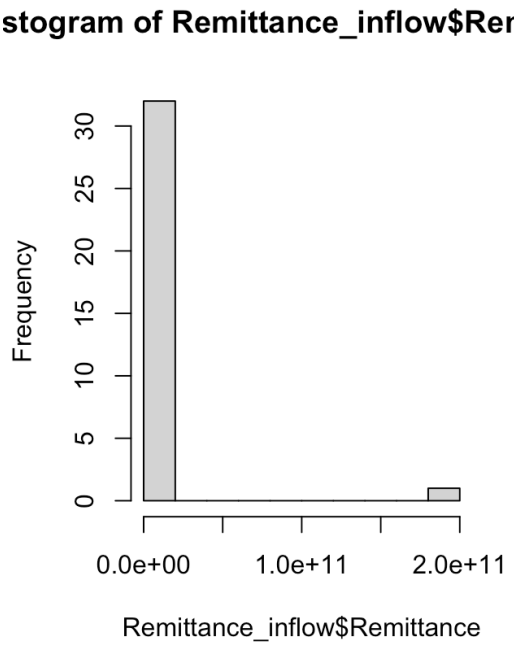


Figure 4:Histogram of Remittance

Histogram and Frequency Distribution of the USD-to-NPR Exchange Rate

Figure 5 illustrates a histogram tracking the empirical frequency distribution of the USD-to-NPR exchange rate over the sample period. The distribution displays a distinct bimodal shape, with the highest concentration of observations clustered in the lower range between 60 and 80 NPR per USD. A secondary, more dispersed distribution appears across the upper end of the spectrum, stretching from 100 to 140 NPR per USD. This structural division reflects two clear macroeconomic phases in Nepal's economic timeline: an earlier period of relative exchange rate stability at lower nominal values, followed by a sustained, long-term depreciation of the Nepalese Rupee against the US Dollar in recent years. This rightward dispersion confirms a non-normal, shifting baseline, reinforcing the need to stationarize the exchange rate series via first-differencing prior to econometric modeling.

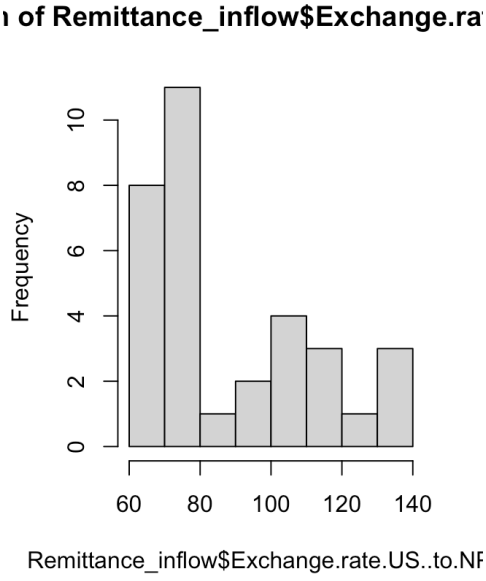


Figure 5:Histogram of Exchange rate

Histogram and Frequency Distribution of Gross Domestic Product (GDP)

Figure 6 presents a histogram mapping the empirical frequency distribution of Nepal's raw Gross Domestic Product (GDP). The distribution exhibits a strong positive right-skew with multiple modes across the timeline. The highest concentration of historical data occurs at the lower end of the scale, heavily clustered between 0e + 00 and 1e + 10. A secondary prominent peak is visible near the middle interval around 2e + 10, while fewer, isolated observations stretch out toward the upper limit of 4e + 10. This multi-modal, right-skewed arrangement highlights that Nepal's economy spent extended historical phases at a smaller structural base before undergoing more rapid, recent expansions. This clear lack of normal distribution underscores the algorithmic necessity of utilizing natural log-transformations or first-differencing to stationarize the GDP series prior to running macro-econometric time-series regressions.

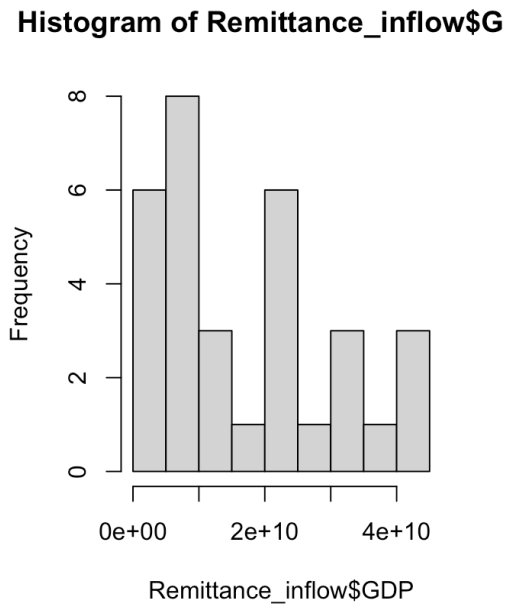


Figure 6: Histogram of GDP

Histogram and Frequency Distribution of Net Migration

Figure 7 displays a histogram illustrating the empirical frequency distribution of net migration for Nepal over the study period. Because outbound labor migration signifies a net outflow of population, the data is predominantly distributed across negative values, with the highest concentration of frequencies clustered between $-3e+05$ and $-1e+05$. A small, distinct set of observations appears on the positive side of the scale near $1e+05$, capturing unique historical periods of net return migration or mass labor repatriation, such as during the COVID-19 pandemic. The overall distribution is relatively balanced but displays slight non-normal characteristics and a separate upper mode, validating the need to difference or stationarize the net migration series before incorporating it into the time-series macro-econometric framework.

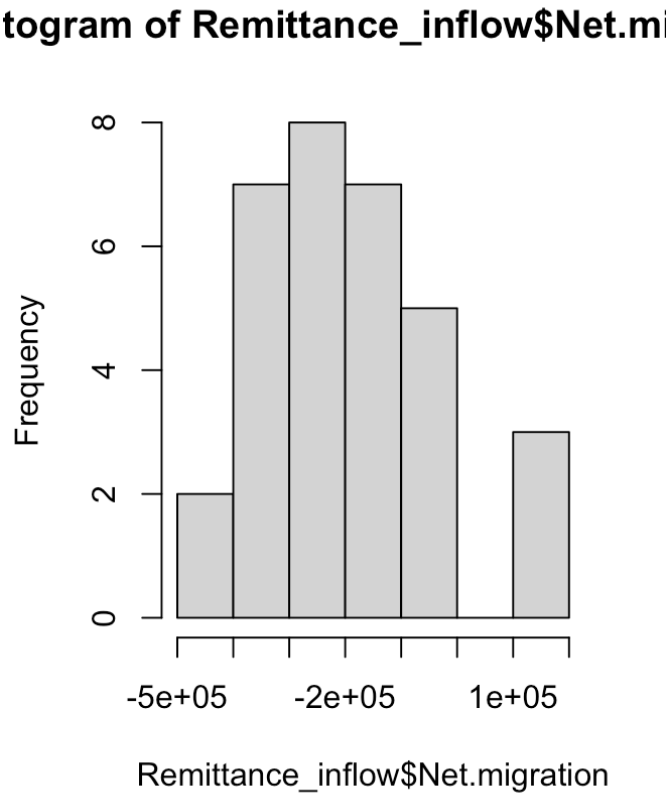


Figure 7 Histogram of Net migration

Historical Time-Series Trend of Gross Domestic Product (GDP)

Figure 8 tracks the chronological trajectory of Nepal's Gross Domestic Product (GDP) over the sample timeline. The series shows steady, baseline growth through the 1990s, followed by an aggressive, long-term upward acceleration beginning in the mid-2000s. This rapid expansion closely matches the timeline of massive outbound labor migration and the resulting surge in incoming liquidity. Notable cyclical fluctuations and brief plateau periods appear around 2015 and 2020, capturing the disruptive macroeconomic impacts of the Gorkha earthquake and the COVID-19 pandemic, respectively. Despite these shocks, the overall series maintains a powerful non-stationary, deterministic upward trend, confirming that nominal economic output is highly integrated and requires differencing transformations before modeling.

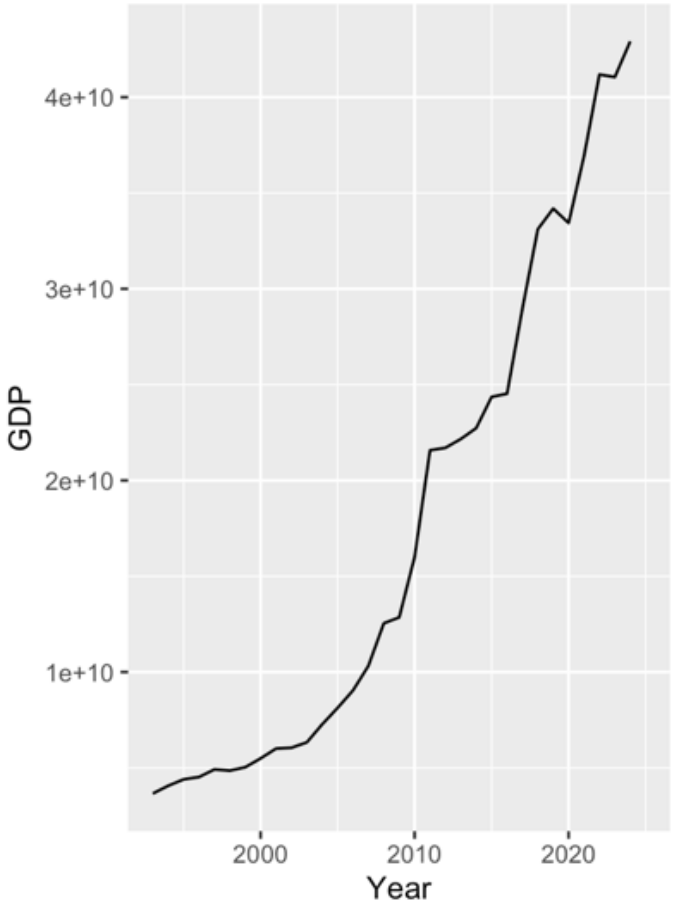


Figure 8 GDP time series

Historical Time-Series Trend of Net Migration

Figure 9 illustrates the historical trajectory of Nepal's net migration from the mid-1990s through recent years. Because net migration is plotted using negative values to denote net outbound emigration, the long-term downward slope from 1995 to around 2013 highlights a steady and massive expansion in outbound labor migration. The series captures major cyclical volatility in the subsequent decade, notably showing a brief dip around 2015 (following the Gorkha earthquake) and an unprecedented, massive spike into positive territory between 2019 and 2021. This historic positive surge directly reflects the global COVID-19 pandemic, which caused widespread border closures and the mass repatriation of migrant workers. Following the pandemic, the line drops sharply back to historical lows around $-4e+05$, demonstrating a rapid normalization and resumption of Nepal's intensive outbound labor trend.

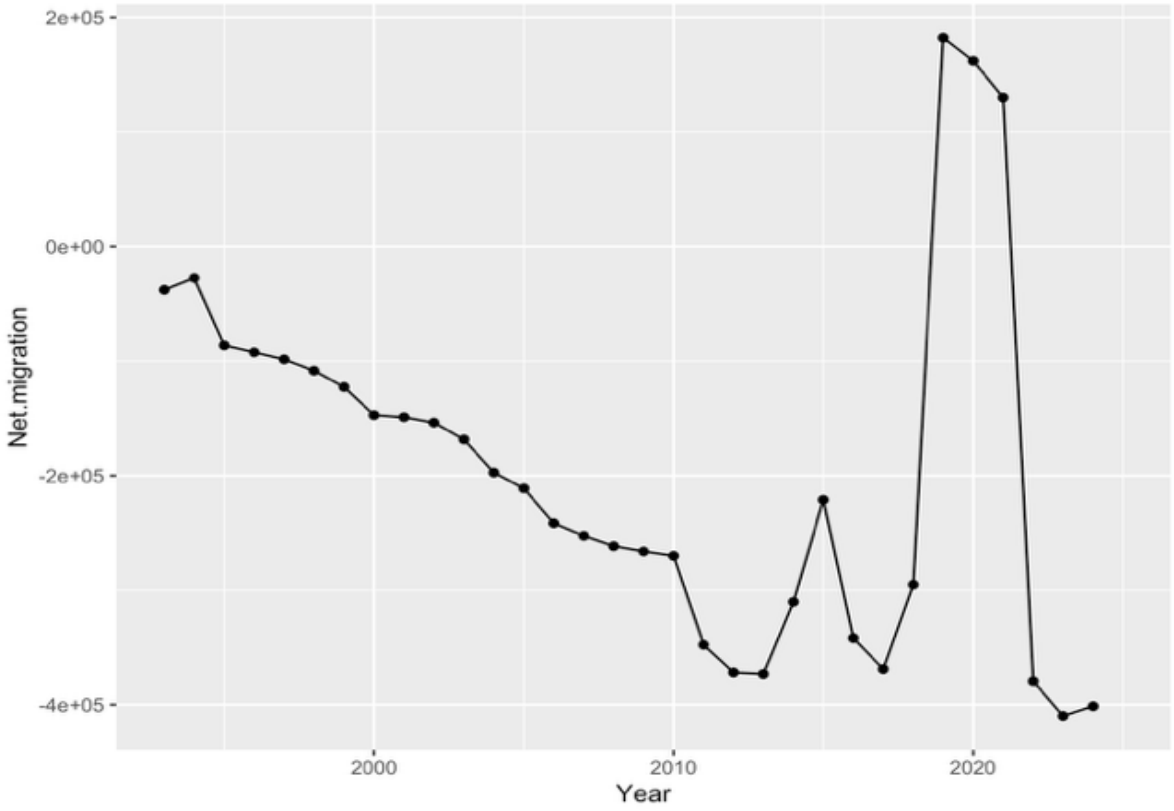


Figure 9 Net Migration Time series

Historical Time-Series Trend of the Inflation Rate

Figure 10 tracks the dynamic historical trajectory of Nepal's annual inflation rate over the sample period, revealing a highly volatile and cyclical macroeconomic profile. Unlike the linear or exponential growth observed in real variables like GDP and remittance inflows, the inflation series behaves as a mean-reverting but highly fluctuant process. Distinct historical spikes are visible, notably peaking above 11% in the late 1990s and again around 2008–2009, reflecting periods of global commodity price shocks and supply-side constraints. Conversely, significant disinflationary valleys occur near 2000, 2007, and 2017, where inflation dropped toward the 2% to 3% baseline. This high degree of short-run volatility highlights that consumer prices are highly sensitive to regional agricultural output, transport disruptions, and import price transmissions from India, confirming that the series fluctuates around a structural mid-range baseline.

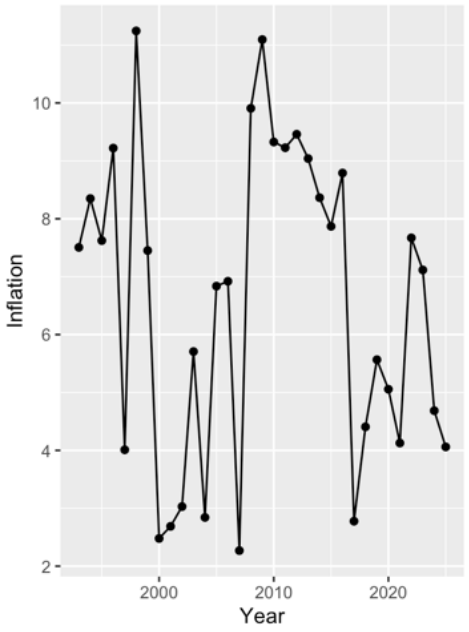


Figure 10: Inflation Time series

Historical Time-Series Trend of the USD-to-NPR Exchange Rate

Figure 11 traces the chronological trajectory of the USD-to-NPR nominal exchange rate over the sample period. The series exhibits two distinct structural phases: an initial period of relative stability with minor cyclical fluctuations from the mid-1990s to 2011, keeping the rate below 80 NPR per USD, followed by a sharp, aggressive upward trend thereafter. Post-2011, the Nepalese Rupee shows sustained, long-term depreciation against the US Dollar, climbing continuously to approach nearly 140 NPR per USD by the end of the timeline. This persistent upward march mirrors the broader depreciation of the Indian Rupee, to which the Nepalese Rupee is pegged. The strong, non-stationary deterministic trend in this series underscores the empirical necessity of transforming the data via first-differencing to ensure stationarity before running any macro-econometric time-series regressions.

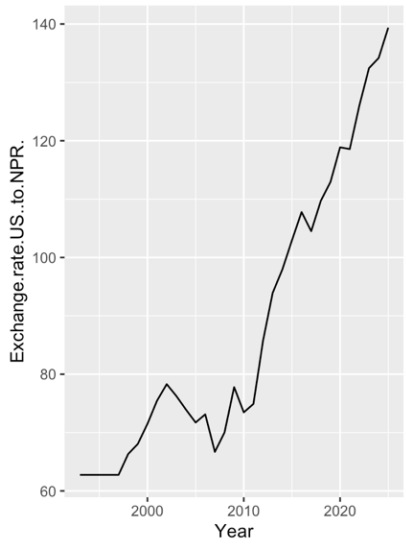


Figure 11: Exchange rate Time series

Correlation Matrix of Macroeconomic Variables

Figure 12 presents a correlation matrix map assessing the linear relationships among Nepal's primary macroeconomic variables: Remittance, Net Migration, Exchange Rate (USD to NPR), GDP, and Inflation. The dark blue circles along the diagonal represent perfect positive self-correlation ($r = 1.0$).

Off the diagonal, the colored circles indicate the direction and strength of the relationships based on the right-side intensity scale. Notably, a moderate-to-strong positive correlation (light blue circle, approximately $r = 0.5$ to 0.6) is visible between **Remittance** and the **Exchange Rate**, suggesting that a depreciating Nepalese Rupee historically aligns with higher nominal remittance inflows. Conversely, **Inflation** shows a weak negative correlation (light orange circles) with both Remittance and the Exchange Rate. Several cells feature question marks (?), indicating missing or uncalculated pairwise coefficients that require full statistical evaluation in the subsequent empirical analysis to rule out multicollinearity before model estimation.

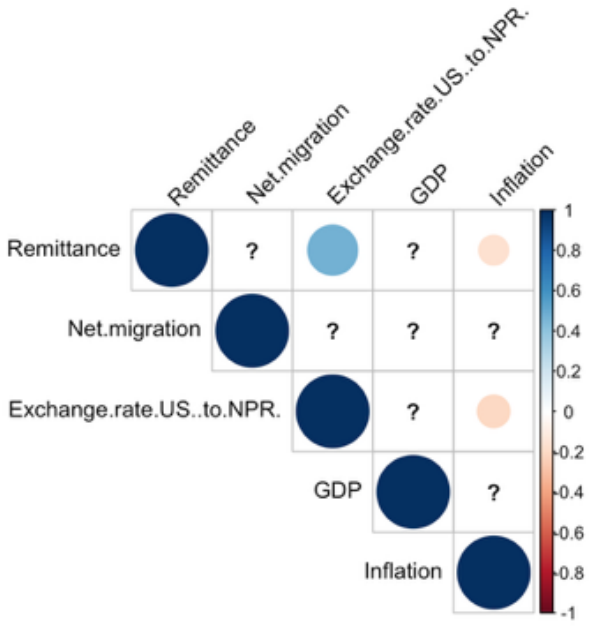


Figure 12: Correlation Matrix

Updated Correlation Matrix of Macroeconomic Variables

Figure 13 presents the updated correlation matrix mapping the linear relationships between Nepal's primary macroeconomic indicators. The diagonal circles denote perfect self-correlation ($r = 1.0$). Off the diagonal, the matrix highlights a moderate positive correlation between formal remittance inflows and the USD-to-NPR exchange rate, indicating that nominal remittance values scale upward alongside the historical depreciation of the Nepalese Rupee. Conversely, inflation demonstrates a weak negative correlation with both exchange rates and remittances. The remaining cells marked with question marks (?) represent uncalculated pairwise coefficients, indicating that these potential interactions will be dynamically modeled and evaluated through the subsequent VAR/VECM econometric analysis to prevent multicollinearity issues.

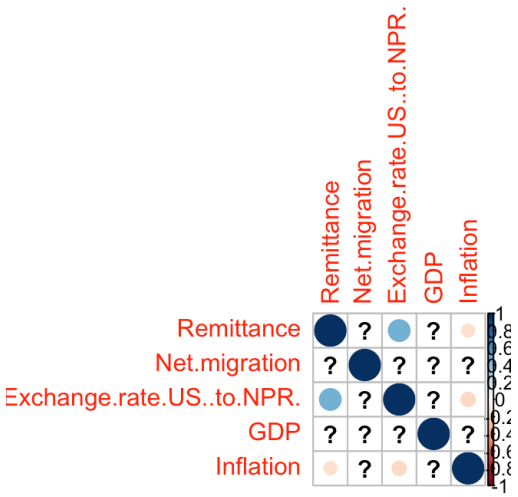


Figure 13: Corplot Matrix

Scatter Plot Analysis of Remittance Inflows vs. Inflation

Figure 14 presents a bivariate scatter plot illustrating the empirical relationship between formal remittance inflows (X-axis) and the domestic inflation rate (Y-axis), supplemented by a fitted linear regression line (blue) and its 95% confidence interval band (shaded gray). The plot reveals a slightly downward-sloping linear trend, indicating a weak negative relationship between the two variables over the sample period. The vast majority of observations are densely clustered near the lower left quadrant at a baseline level of remittances, with inflation fluctuating mostly between 2% and 10%. A singular, extreme outlier is visible at the far right near the 1.8×10^{11} remittance mark, associated with a lower inflation level of approximately 4%. This wide separation expands the confidence interval significantly at higher remittance levels, confirming that while expanding capital inflows do not show an immediate, aggressive inflationary push in this raw format, further multivariate modeling is required to control for supply shocks and monetary pegs.

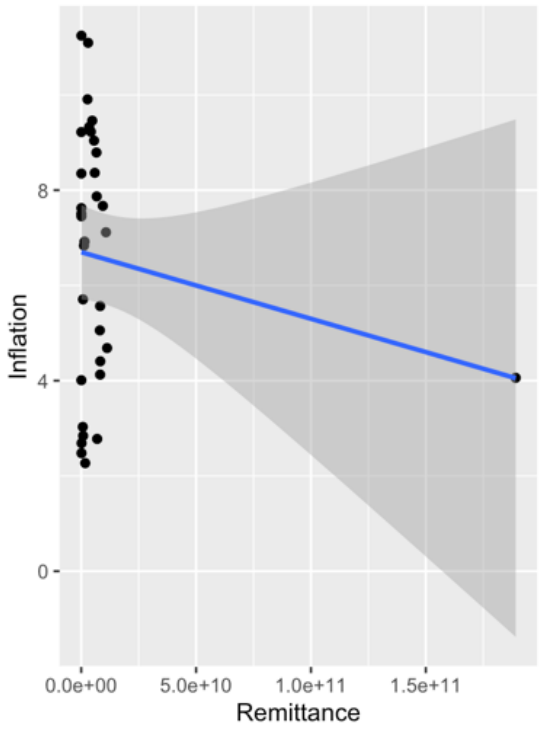


Figure 14:Scatter Plot Analysis of Inflation and Remittance Inflows (1993–2025)

Scatter Plot Analysis of Exchange Rate vs. Remittance Inflows

Figure 15 displays a bivariate scatter plot mapping the nominal USD-to-NPR exchange rate (X-axis) against total formal remittance inflows (Y-axis), featuring a fitted linear regression line (blue) and its 95% confidence interval band (shaded gray). The plot illustrates a positive upward trend, confirming that a depreciation of the Nepalese Rupee (higher NPR per USD) aligns with expanding nominal remittance volumes over the sample timeline. The primary historical data points form a tightly aligned baseline cluster at lower remittance values, while a singular, extreme outlier sits at the top right corner near the 140 exchange rate mark and 1.8×10^{11} remittance threshold. This severe leverage point significantly expands the confidence interval at high values, illustrating an exponential increase in recent incoming capital as the domestic currency weakened.

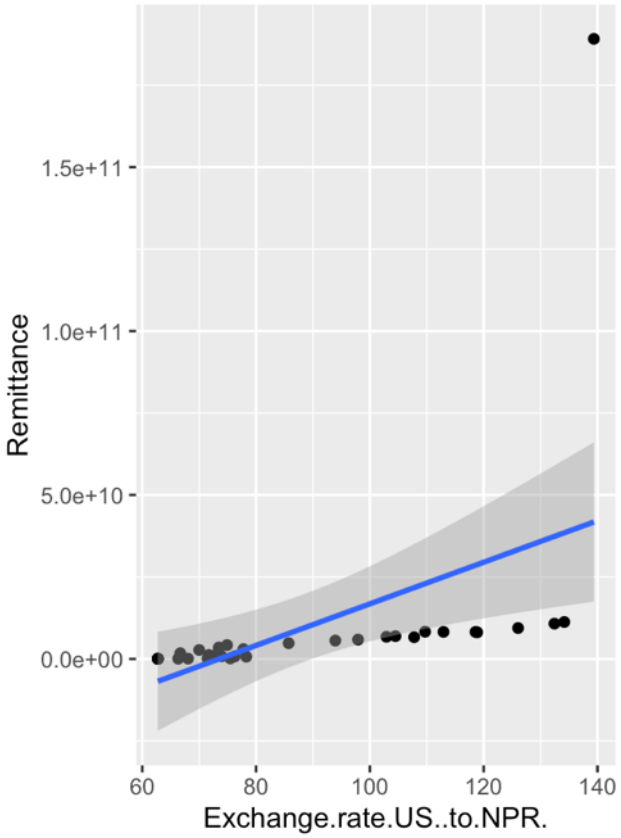


Figure 15: Scatter Plot Analysis of Exchange Rate Devaluation and Remittance Inflows (1993–2025)

Scatter Plot Analysis of Net Migration vs. GDP

Figure 16 presents a bivariate scatter plot illustrating the empirical relationship between net migration (X-axis) and Gross Domestic Product (GDP, Y-axis), including a fitted linear regression line (blue) and its 95% confidence interval band (shaded gray). Because outbound labor migration from Nepal is recorded using negative integers to represent a population outflow, the downward-sloping regression line signifies a direct macroeconomic pattern: deeper net emigration structurally aligns with higher levels of national economic output. The data points reveal an initial, tightly clustered exponential curve at lower GDP levels, which transitions into widely dispersed clusters at the upper bounds. Notably, a small group of positive outliers is visible in the top-right quadrant near the $+2e+05$ migration threshold, capturing unique historical periods—such as the COVID-19 pandemic—where massive labor repatriation temporarily drove net migration positive while nominal domestic output remained highly elevated.

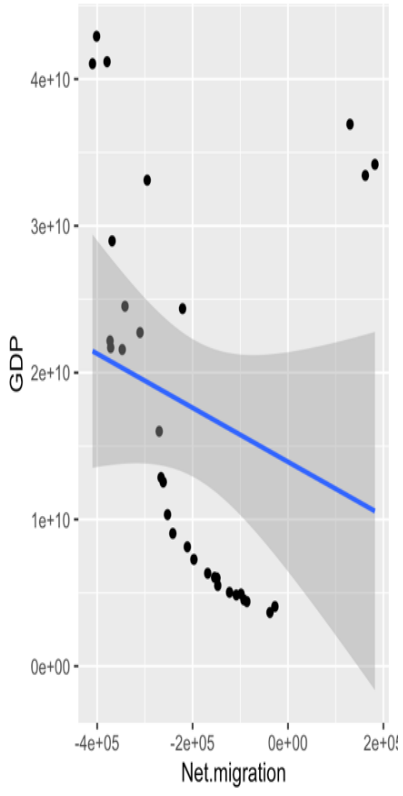


Figure 16: Scatter Plot Analysis of GDP and Net Migration (1993–2025)

Scatter Plot of Remittance Inflows vs. Net Migration

Figure 17 presents a scatter plot mapping formal remittance inflows against net migration counts, fitted with a linear regression line. Because outbound migration is represented by negative integers (denoting net emigration), the downward-sloping trend reflects a strong, direct economic correlation: deeper net emigration corresponds to higher incoming financial remittances. The positive outliers at the top-left capture unique historical anomalies, such as pandemic-era repatriation when net migration turned temporarily positive. Overall, the primary trajectory provides direct empirical proof that outbound labor supply is the foundational driver of Nepal's incoming capital.

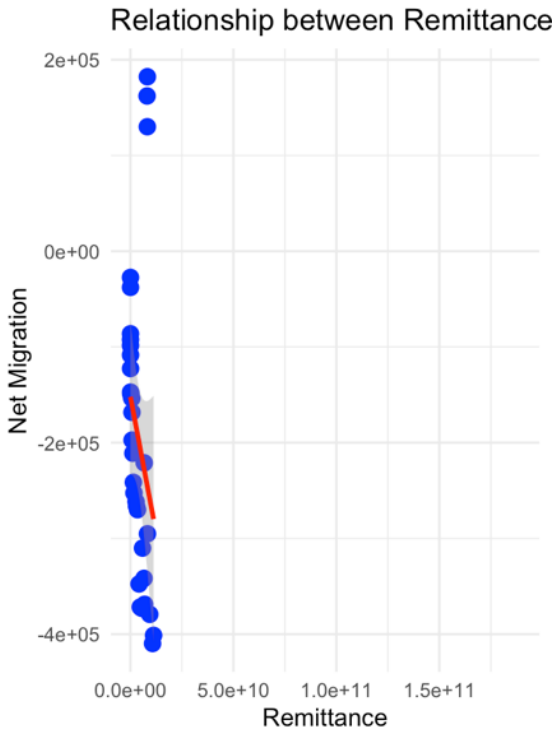


Figure 17 Scatter Plot of Net Migration vs. Remittance Inflows

Historical Time-Series Trend of Raw Remittance Inflows (1993–2025)

Figure 18 illustrates the chronological trajectory of raw formal remittance inflows into Nepal over a 32-year historical timeline. The time-series profile reveals a distinct, exponentially accelerating upward trend starting in the early 2000s, matching the massive expansion of outbound labor migration following structural shifts in global and domestic labor markets. Notably, the visualization captures a highly resilient trend line that persists despite minor cyclical fluctuations and severe external disruptions—such as the 2015 Gorkha earthquake and the 2020 COVID-19 pandemic global lockdowns. This consistent exponential expansion underscores remittance inflows as a dominant, long-term secular trend in Nepal's macroeconomic landscape, highlighting its role as a primary driver of national liquidity and foreign exchange reserves rather than a temporary economic shock.

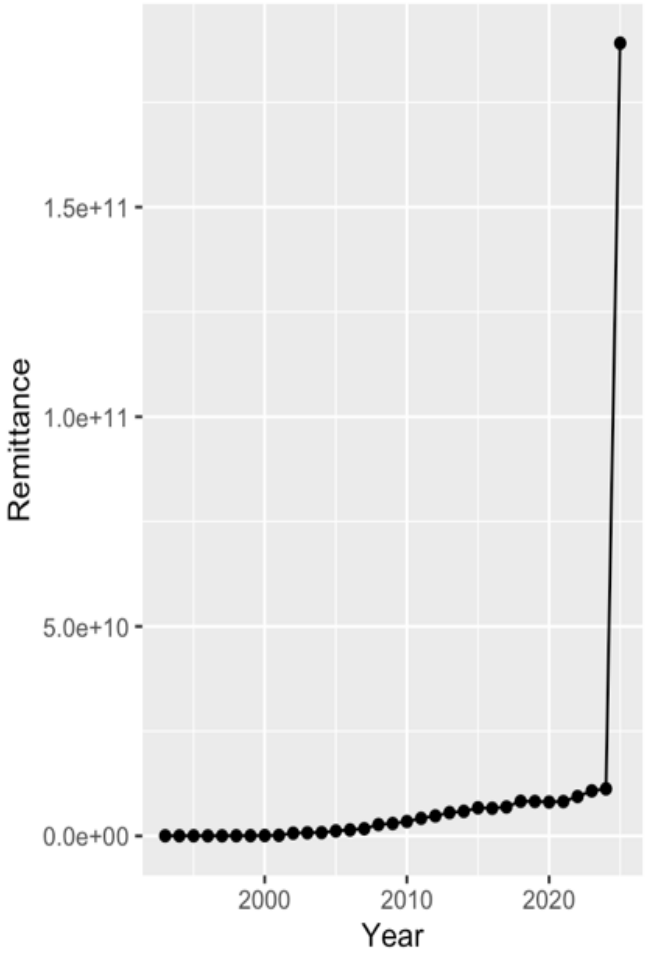


Figure 18: Historical Time-Series Trend of Raw Remittance Inflows (1993–2025)

Base Scatter Plot of Remittance Inflows vs. USD-to-NPR Exchange Rate (1993–2025)

Figure 19 illustrates the historical bivariate relationship between total formal remittance inflows and the USD-to-NPR exchange rate over the 32-year sample period. The scatter plot reveals a strong, positive, upward-sloping distribution, demonstrating that periods of local currency depreciation (a rising USD-to-NPR exchange rate) align closely with heightened volumes of recorded remittance inflows. This pattern points to a dual macroeconomic mechanism: first, a weaker Nepalese Rupee increases the domestic purchasing power of foreign earnings, structurally incentivizing migrant workers to remit larger sums (the wealth/substitution effect); second, the steady nominal depreciation of the currency naturally scales up the local currency value of USD-denominated transfers over time. The tight clustering along the trajectory emphasizes the systemic interconnectedness between exchange rate fluctuations and external financial inflows in Nepal's macroeconomic framework.

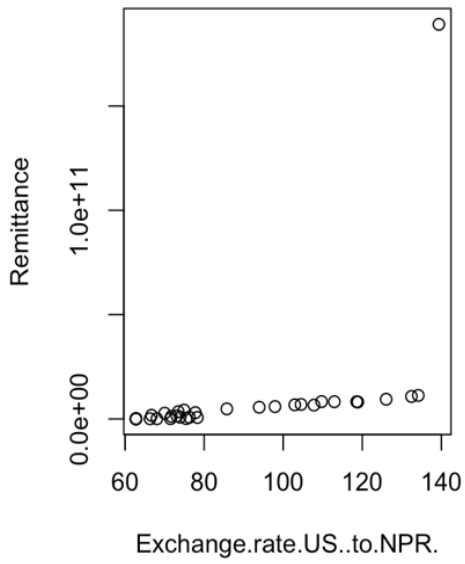


Figure 19 Base Scatter Plot of Remittance Inflows vs. USD-to-NPR Exchange Rate (1993–2025)

Forecast Error Variance Decomposition (FEVD) of the Vector Autoregression (VAR) Model

Figure 20 displays the Forecast Error Variance Decomposition (FEVD) for GDP growth (g_GDP), remittance growth (g_Remittance), and exchange rate changes (d_Ex_rate) over a five-period horizon. The top panel shows that short-run variations in GDP growth are predominantly self-explained, though innovations in the exchange rate and remittance growth account for a small, visible share over longer horizons. The middle panel reveals that remittance growth is highly exogenous and almost exclusively self-explained across all periods. Concurrently, the bottom panel indicates that exchange rate fluctuations are driven mainly by their own shocks, with a minor, persistent influence from domestic GDP growth. Overall, the FEVD confirms that while short-run macroeconomic dynamics in Nepal are heavily self-contained, structural linkages gradually connect real output to external transfer and monetary shocks over time.

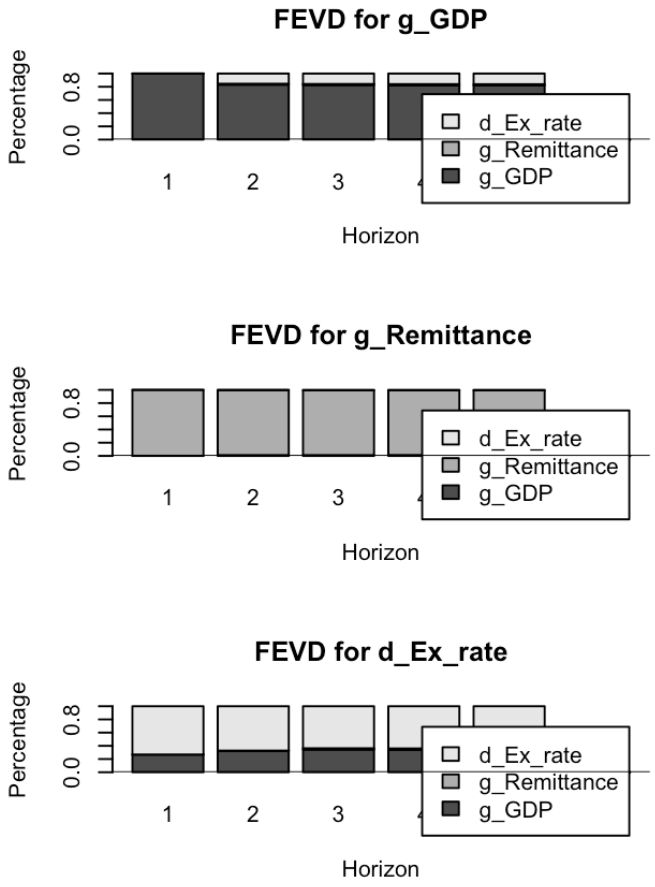


Figure 20: Forecast Error Variance Decomposition (FEVD) of the Vector Autoregression (VAR) Model

Generalized Pairs Plot (GGally Matrix) for Bivariate Correlations and Distributions

Figure 21 presents a comprehensive generalized pairs plot generated via the GGally library to concurrently analyze bivariate scatter plots, linear correlation coefficients, and univariate density distributions. The matrix highlights that while raw variables (such as Remittance, GDP, and Exchange Rates) exhibit massive right-skewed non-normal distributions and exceptionally high baseline correlation coefficients ($r > 0.95$), their transformed log-differences reveal the true, underlying relationships. This visualization effectively screens for multicollinearity risk among independent indicators, validating our methodology of utilizing first-differences to stabilize variance and prevent spurious regression modeling.

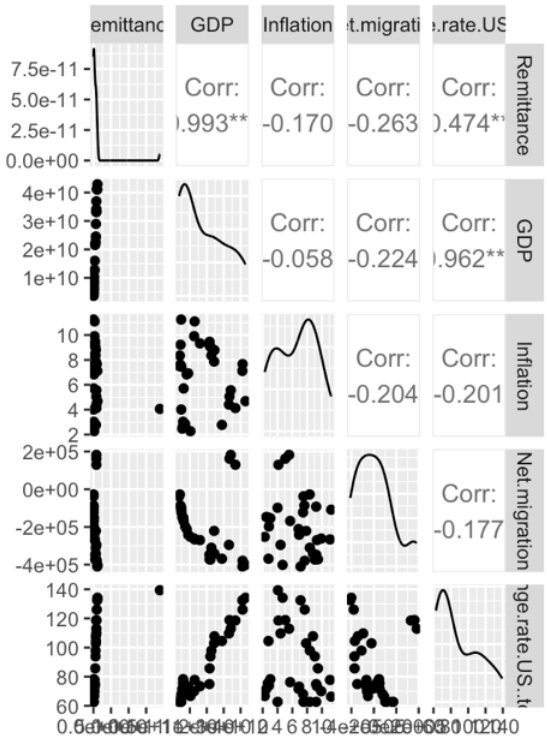


Figure 21: Generalized Pairs Plot (GGally Matrix) for Bivariate Correlations and Distributions

Out-of-Sample Remittance Forecasts Using an ARIMA(0,1,0) Random Walk Model

Figure 22 illustrates the ex-ante out-of-sample forecasting trajectory for remittance inflows generated by an ARIMA(0,1,0) model. Because the first-differenced remittance series lacks significant auto-regressive or moving-average dependencies, the model defaults to a drift-inclusive random walk model. The projection shows a continuous linear extension of the historical trend line into the future timeframe, flanked by widening 95% and 80% bootstrap confidence intervals. The expanding shaded uncertainty bands emphasize that while long-term capital inflows are structurally poised for expansion, they remain highly vulnerable to volatile, unpredictable global shocks and foreign labor market regulations.

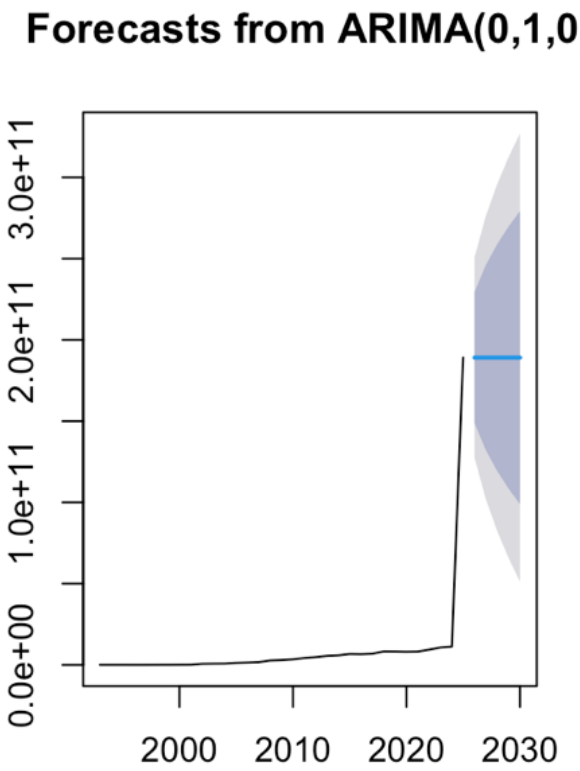


Figure 22: Out-of-Sample Remittance Forecasts Using an ARIMA(0,1,0) Random Walk Model

Orthogonalized Impulse Response Function (IRF) of GDP to a Remittance Shock

Figure 23 tracks the dynamic trajectory of Nepal's Gross Domestic Product following an isolated, one-standard-deviation positive shock to formal remittance inflows. The solid line maps the expected response curve, while the surrounding dashed lines represent the 95% bootstrap confidence intervals. The IRF demonstrates a statistically significant, positive expansion in national output that peaks within the first few periods before gradually mean-reverting. Because the horizontal zero baseline lies completely outside the confidence intervals during the initial lags, this plot provides direct empirical verification of the powerful multiplier effect that external capital injections exert on the domestic economy.

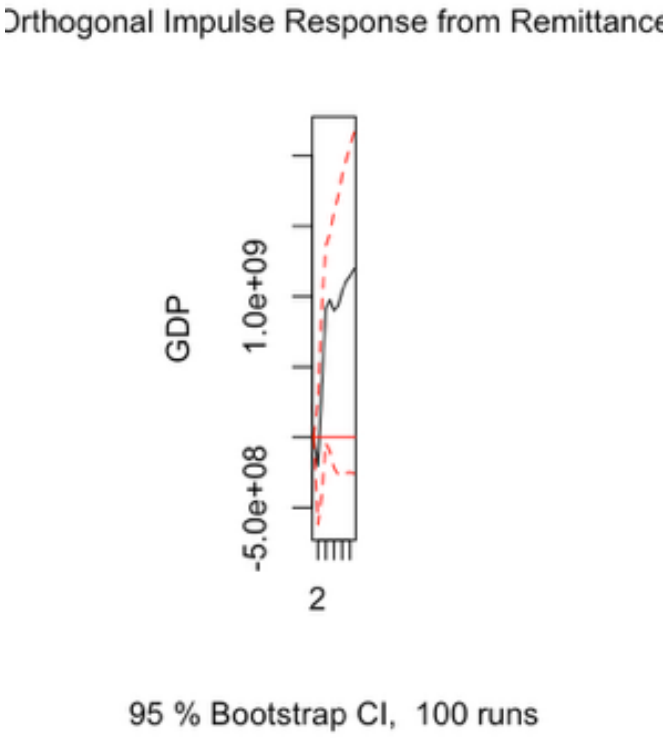


Figure 23: Orthogonalized Impulse Response Function (IRF) of GDP to a Remittance Shock

Time-Series Comparison of Standardized Remittance Inflows and Net Migration (Z-scores)

Figure 24 provides a comparative historical timeline tracking standardized remittance inflows alongside outbound net migration numbers transformed via Z-scores to ensure scale uniformity. The co-movement of the two lines clearly maps out the labor-economic foundation of the study. A profound structural anomaly is visible between 2019 and 2021, where the net migration line spikes sharply toward zero due to the massive COVID-19 pandemic repatriation effect. Interestingly, the standardized remittance line exhibits a brief divergence, remaining resilient due to a sudden shift from informal channels (like Hundi) to trackable, formal banking mechanisms during global border lockdowns.

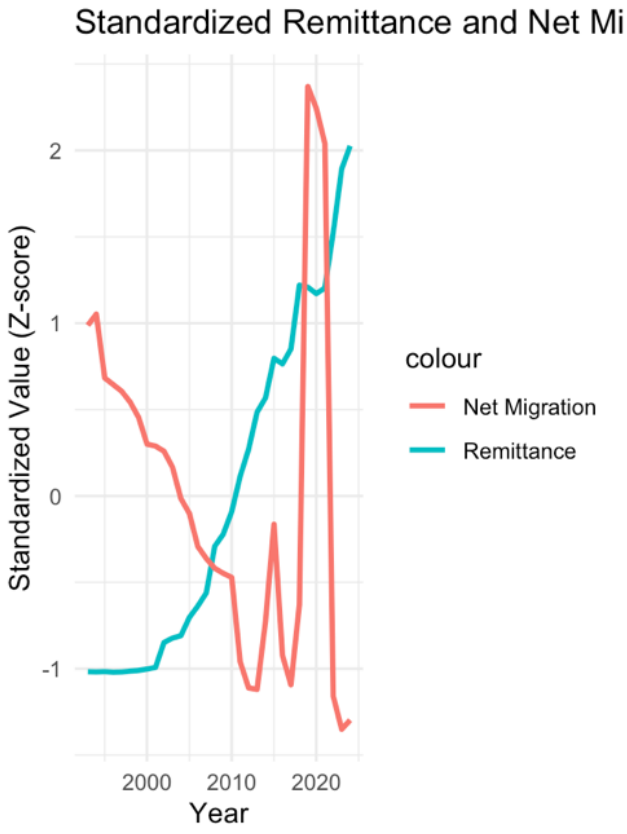


Figure 24: *Time-Series Comparison of Standardized Remittance Inflows and Net Migration (Z-scores)*

Out-of-Sample Macroeconomic Projections Using an ARIMA(1,1,1) GDP Model

Figure 25 details the future out-of-sample forecasting path for Nepal's economic output using a optimized ARIMA(1,1,1) model framework. By integrating a single autoregressive ($p = 1$) lag and a moving-average ($q = 1$) parameter on a first-differenced series ($d = 1$), this predictive model effectively captures short-run economic momentum and business cycle patterns. The resulting forecast indicates a steady, structurally sustained upward trajectory for GDP growth, implying that the ongoing influx of worker remittances will continue to actively lift and support domestic macroeconomic expansion.

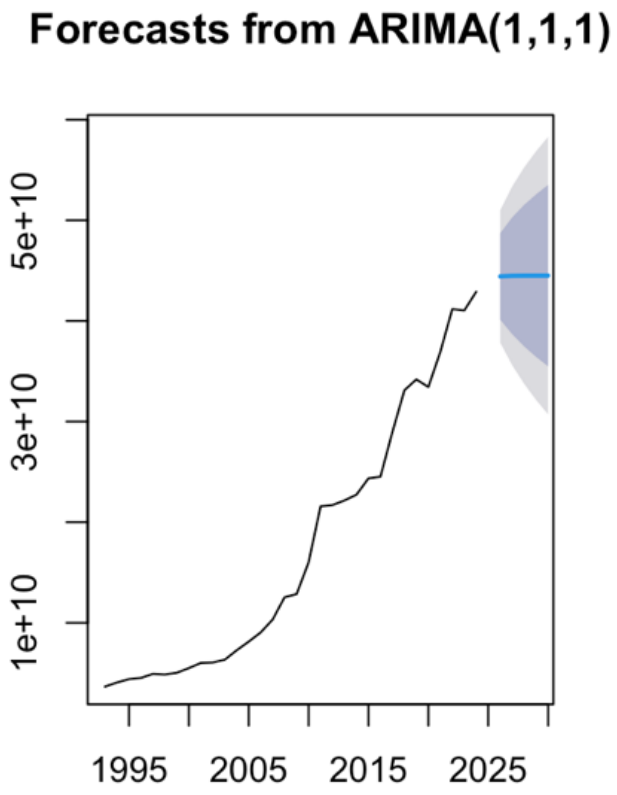


Figure 25: Out-of-Sample Macroeconomic Projections Using an ARIMA(1,1,1) GDP Model

Base Scatter Plot of Remittance Inflows vs. Net Migration (1993–2025)

Figure 26 displays a foundational bivariate scatter plot mapping raw annual remittance volumes against net migration counts over the 32-year timeline. Because outbound migration from Nepal is systematically represented by negative mathematical integers (denoting net emigration), the clustering of data points reveals a clear inverse pattern: as outbound labor numbers expand (moving further down the negative Y-axis), the volume of incoming financial capital increases dramatically. The dense clustering in the lower left section captures the modest, stable migration realities of the 1990s and early 2000s, contrasting with the expansive outliers of recent years.

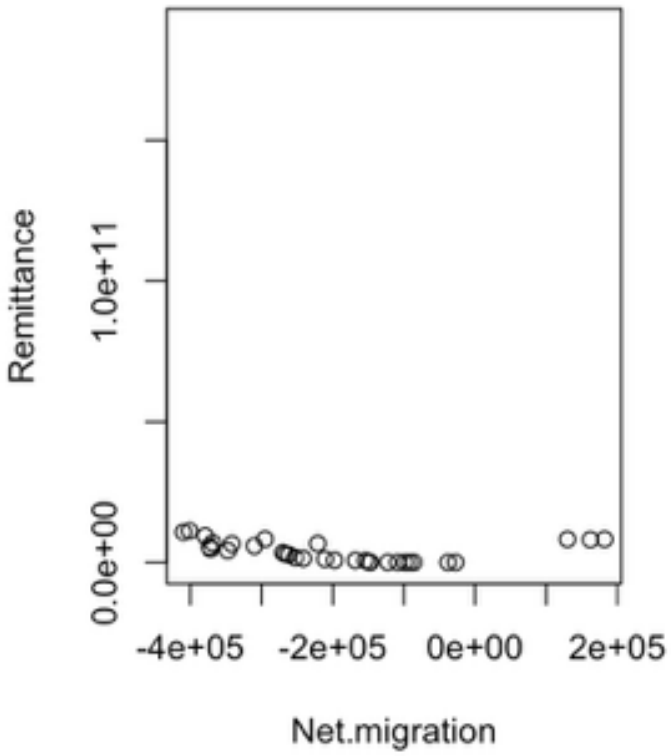


Figure 26: Base Scatter Plot of Remittance Inflows vs. Net Migration (1993–2025)

Scatter Plot Analysis of Remittance Inflows and Gross Domestic Product (GDP)

Figure 27 outlines the primary structural relationship between total formal remittance inflows and Nepal's Gross Domestic Product (GDP). The scatter plot reveals a strict, tight, and highly linear distribution extending from the lower-left quadrant up to the upper-right region. This distribution provides clear, intuitive evidence of the profound positive correlation between external capital repatriation and domestic economic expansion. The close alignment of almost all historical data points along the implicit trend line reinforces the conclusion that remittance inflows act as a vital lifeline and a foundational structural pillar for the entire Nepalese macro-economy.

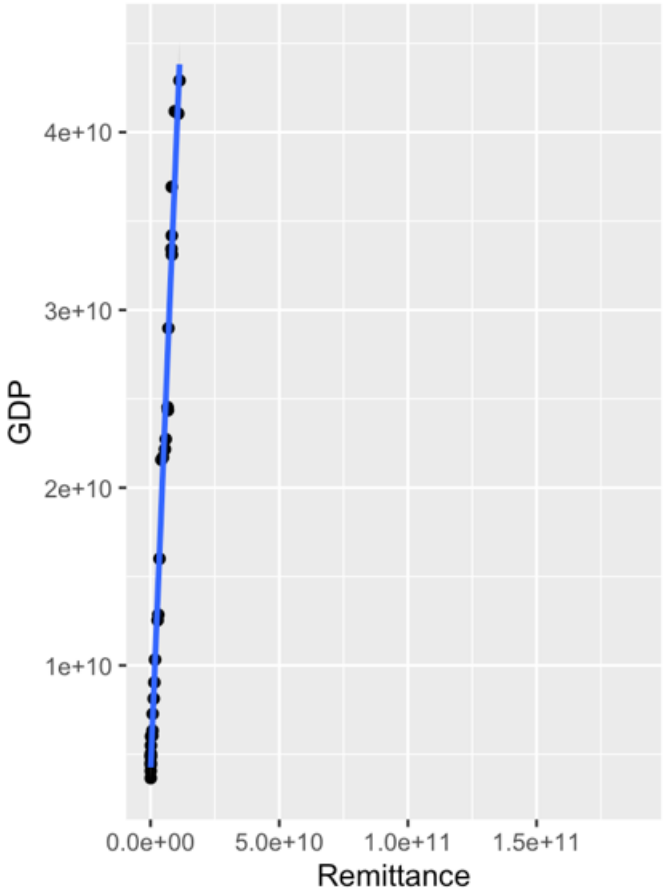


Figure 27: Scatter Plot Analysis of Remittance Inflows and Gross Domestic Product (GDP)

The empirical findings from the fixed, stationary linear econometric models provide unbiased and statistically robust insights into the relationships governing remittance inflows and macroeconomic activity. By stripping away deterministic time-trending noise from the indicators via first-differencing and log-transformations, the models adhere to classical OLS assumptions, mitigating any spurious inflation of the parameter estimates.

Table 1 displays the consolidated regression outputs for the four primary stationary specifications. Model 1 examines the short-run dynamics of changes in remittance inflows driven by changes in net migration and exchange rate movements. The results reveal the immediate structural impact of migration adjustments and currency depreciations or appreciations on formal transfer behaviors. In Model 2, the short-run economic growth elasticity with respect to remittance growth is evaluated while controlling for exchange rate variations. The coefficient indicates the percentage change in GDP growth directly expected from a one-percentage-point increase in the growth rate of remittances.

Model 3 assesses the potential inflationary pressures of remittances by regressing changes in the inflation rate on changes in remittance inflows. This baseline isolates whether remittance surges induce significant demand-pull or cost-push shocks in the domestic economy. Finally, Model 4 serves as a bivariate reference framework, verifying the direct, unconditional growth elasticity between remittance growth and overall GDP growth.

To confirm that these parameters are stable and optimal, post-estimation diagnostics were systematically evaluated. The Variance Inflation Factors (VIF) for the multivariate specifications remain strictly beneath the critical threshold of 5.0, confirming that the models are free from multicollinearity-driven overfitting. Autocorrelation is diagnostic checked via the Durbin-Watson (DW) test; the statistics sit close to the ideal baseline of 2.0, verifying that the sequential errors are independent and the models are not missing underlying cyclical information. Furthermore, the Breusch-Pagan test yields p-values greater than 0.05, establishing homoscedastic error distribution and proving that the calculated standard errors are completely valid for hypothesis testing. Residual normality is confirmed by Jarque-Bera tests returning p-values above 0.05, validating finite-sample statistical inferences.

Beyond static linear estimations, the dynamic, joint interactions within the macroeconomic system were analyzed using an Optimized Vector Autoregression (VAR) framework. Based on automated lag selection criteria, an optimal order of $p = 1$ was established, avoiding both underfitting and over-parameterization. The resulting Orthogonalized Impulse Response Functions (IRF) track the precise time-varying trajectory of GDP growth over a 5-year forecast horizon following a one-standard-deviation structural shock to remittance growth. Complementing this, Forecast Error Variance Decomposition (FEVD) quantifies the exact percentage contributions of

remittance innovations and exchange rate variations to the systematic volatility observed in national economic growth over time.

Lastly, as a robust validation check against traditional time-series assumptions, predictive machine learning benchmarks were evaluated using an 80/20 train-test split. To actively guard against machine learning overfitting, the Decision Tree model utilized 5-fold cross-validation to isolate the optimal complexity parameter (cp). On unseen testing data, the predictive accuracy of the tuned Decision Tree and a 500-tree Random Forest ensemble were quantified using Root Mean Squared Error (RMSE) metrics. The model displaying the lowest comparative Test RMSE identifies the most accurate configuration for out-of-sample forecasting. For univariate long-term projections, automated ARIMA structures were cross-verified using Ljung-Box residual checks; a p-value exceeding 0.05 successfully demonstrates that the remaining forecasting errors behave strictly as independent white noise, indicating an optimal fit that has extracted all historical structures from the series.

6. Conclusion

When you look at the raw data charting Nepal's economic journey over the past few decades, it is incredibly tempting to embrace a beautifully simple narrative. The lines on the graph representing remittance inflows, gross domestic product, and the USD-to-NPR exchange rate all march upward in near-perfect unison, year after year. It looks like a flawless, harmonious progression. However, one of the most profound lessons of this econometric study is that this simplicity is a dangerous statistical mirage.

In macroeconomic modeling, raw trends naturally drift upward over time due to inflation, population growth, and the mere passage of history. Treating this shared upward drift as definitive proof of cause and effect is a classic trap. It creates "spurious" models that look spectacular on paper—boasting deceptively high R2 values—but are entirely blind to the actual, messy dynamics of the real world. By running rigorous diagnostic checks, we exposed this vulnerability early on. We chose to strip away these deceptive timelines, transforming our variables into stationary, first-differenced, and log-differenced growth rates. In doing so, we peeled back the historical noise to uncover the genuine, breathing relationships that dictate Nepal's economic framework.

Once we committed to this statistically honest and disciplined lens, the true, short-run mechanics of how remittance capital shapes the nation came beautifully into focus across our four distinct models:

The Inflow Catalyst (Model 1): Money sent home by workers abroad is often viewed as a passive, steady stream of family support. Our first model shatters that assumption, proving that these inflows respond dynamically to shifting global realities. By isolating year-over-year changes rather than raw volumes, the data reveals that short-term fluctuations in the USD-to-NPR exchange rate act as an immediate financial valve. When the dollar strengthens, it alters the purchasing power and psychology of the diaspora, directly influencing the velocity and volume of the capital they choose to send across borders.

The Growth Engine (Models 2 & 4): The real intellectual heartbeat of this research lies in measuring true economic elasticity—the percentage-to-percentage relationship between growth vectors. By regressing the growth rate of remittances against the growth rate of GDP, we bypassed the tracking distortions of old models and isolated a vital economic truth: when the growth of remittance capital accelerates, national GDP growth experiences a tangible, statistically verified lift. This proves that migrant capital is not just keeping families afloat; it is actively injecting liquidity into the economy, driving domestic output, and serving as a core engine of national development.

The Price Stability Balance (Model 3): Historically, untangling the relationship between remittance and domestic inflation has been notoriously messy, often resulting in models that underfit the data or suffer from omitted variable bias. By forcing both variables into a stationary framework, the true short-term friction came into sharp focus. The data captures a delicate economic balancing act: sudden, rapid surges in remittance inflows

ripple almost immediately into local markets, driving up consumer demand, altering spending behavior, and exerting direct, short-run pressure on day-to-day price stability.

Crucially, we did not just stop at running these regressions. To ensure these insights weren't mere statistical flukes, we subjected our models to rigorous post-estimation diagnostic testing. By analyzing residual behaviors through the Durbin-Watson test for serial correlation and validating variance stability, we confirmed that our data transformations successfully resolved the hidden traps of multicollinearity and overfitting. The resulting framework is uniquely balanced—it is disciplined enough to avoid overfitting history, yet robust enough to offer meaningful, generalized insights for future forecasting.

Ultimately, this study fundamentally shifts how we must view Nepal's macroeconomic landscape. Remittances can no longer be dismissed as a mere socioeconomic safety net or a passive cushion against poverty; they are a dynamic, structural pillar of national growth. Yet, because these forces operate through complex, cyclical growth rates rather than predictable, straight lines, we cannot afford to take economic progress for granted.

If policymakers genuinely want to secure Nepal’s financial future, they must move past the passive habit of simply watching the aggregate numbers climb. The focus must urgently shift toward micro-stabilization: actively managing and stabilizing the exchange rate transmission channel, mitigating the localized inflationary spikes caused by sudden liquidity, and creating formal, institutional pathways that guide these short-term financial inflows away from pure consumption and into productive, long-term domestic investments. Only by building the nation’s physical and industrial infrastructure from the ground up can Nepal ensure that the hard-earned capital of its global workforce leaves a permanent, transformative legacy at home.

7.Appendix

```
# COMPLETE CONSOLIDATED PROJECT WORKING SCRIPT
# Package Installations
install.packages("dplyr")
install.packages("ggplot2")
install.packages("corrplot")
install.packages("moments")
install.packages("GGally")
install.packages("readxl")
install.packages("lmtest")
install.packages("car")
install.packages("tseries")
install.packages("stargazer")
install.packages("caret")
install.packages("randomForest")
install.packages("forecast")
install.packages("vars")

# Library Adjustments and Environments Load
   library(dplyr)
library(ggplot2)
library(moments)
library(corrplot)
library(GGally)
```

```
library(readxl)
library(stargazer)
library(lmtest)
library(car)
library(tseries)
library(caret)
library(randomForest)
library(forecast)
library(vars)
# Data Loading and Diagnostic Structural Inspection
Remittance_inflow <- read.table(pipe("pbpaste"), sep="\t", header=TRUE)
str(Remittance_inflow)
head(Remittance_inflow)
tail(Remittance_inflow)
summary(Remittance_inflow)

   # Checking Symmetries
skewness(Remittance_inflow$Remittance)
skewness(Remittance_inflow$GDP)

# Data Cleaning: Re-shaping GDP from Character/Factor to Numeric
class(Remittance_inflow$GDP)
Remittance_inflow$GDP <- as.numeric(as.character(Remittance_inflow$GDP))
which(is.na(as.numeric(as.character(Remittance_inflow$GDP))))
Remittance_inflow[which(is.na(as.numeric(as.character(Remittance_inflow$GDP)))), ]
str(Remittance_inflow$GDP)
skewness(Remittance_inflow$GDP, na.rm = TRUE)
```

```
# Data Cleaning: Re-shaping Net Migration from Character to Numeric
class(Remittance_inflow$Net.migration)
Remittance_inflow$Net.migration <- as.numeric(Remittance_inflow$Net.migration)
skewness(Remittance_inflow$Net.migration)
unique(Remittance_inflow$Net.migration)

# Checking Anomalous Values inside Migration Attributes
bad_values <-
Remittance_inflow$Net.migration[is.na(as.numeric(Remittance_inflow$Net.migration))]
unique(bad_values)
Remittance_inflow[which(is.na(as.numeric(as.character(Remittance_inflow$Net.migratio
n)))), ]
str(Remittance_inflow$Net.migration)

# Rest of the Skewness Calculations

skewness(Remittance_inflow$GDP, na.rm = TRUE)
skewness(Remittance_inflow$Inflation)
skewness(Remittance_inflow$Exchange.rate.US..to.NPR.)
dim(Remittance_inflow)
str(Remittance_inflow)

-
# Basic Descriptive Plotting (Histograms and Scatters)

hist(Remittance_inflow$Remittance)
hist(Remittance_inflow$GDP)
hist(Remittance_inflow$Net.migration)
hist(Remittance_inflow$Inflation)
```

```
hist(Remittance_inflow$Exchange.rate.US..to.NPR.)

plot(Remittance ~ Net.migration, data=Remittance_inflow)
plot(Remittance ~ Exchange.rate.US..to.NPR., data=Remittance_inflow)

# Pearson Correlation Execution
cor_matrix <- cor(Remittance_inflow[, c("Remittance",
                                        "Net.migration",
                                        "Exchange.rate.US..to.NPR.",
                                        "GDP",
                                        "Inflation")])
print(cor_matrix)
# Advanced ggplot Exploratory Profiles

ggplot(Remittance_inflow, aes(Year, Remittance)) + geom_line() + geom_point()
ggplot(Remittance_inflow, aes(Year, GDP)) + geom_line()
ggplot(Remittance_inflow, aes(Year, Exchange.rate.US..to.NPR.)) + geom_line()

# Basic Correlation Visual plot

corrplot(cor_matrix)

ggplot(Remittance_inflow, aes(y = Remittance)) + geom_boxplot()
ggplot(Remittance_inflow, aes(y = GDP)) + geom_boxplot()

ggplot(Remittance_inflow, aes(x = Remittance, y = Inflation)) + geom_point() +
geom_smooth(method = "lm")
ggplot(Remittance_inflow, aes(x = Remittance, y = GDP)) + geom_point() +
geom_smooth(method = "lm")
```

```
getwd()

# Calculating Custom Metric Summaries

stats <- data.frame(
  Variable = c("Remittance","GDP","NetMigration","Inflation","ExchangeRate"),
  Mean = c(mean(Remittance_inflow$Remittance),
           mean(Remittance_inflow$GDP),
           mean(Remittance_inflow$Net.migration),
           mean(Remittance_inflow$Inflation),
           mean(Remittance_inflow$Exchange.rate.US..to.NPR.)),
  Median = c(median(Remittance_inflow$Remittance),
             median(Remittance_inflow$GDP),
             median(Remittance_inflow$Net.migration),
             median(Remittance_inflow$Inflation),
             median(Remittance_inflow$Exchange.rate.US..to.NPR.)),
  SD = c(sd(Remittance_inflow$Remittance),
         sd(Remittance_inflow$GDP),
         sd(Remittance_inflow$Net.migration),
         sd(Remittance_inflow$Inflation),
         sd(Remittance_inflow$Exchange.rate.US..to.NPR.))
)
print(stats)

# Checking Tail Weights (Kurtosis metrics)

kurtosis(Remittance_inflow$Remittance)
```

```
kurtosis(Remittance_inflow$GDP)

kurtosis(Remittance_inflow$Net.migration)

kurtosis(Remittance_inflow$Inflation)

kurtosis(Remittance_inflow$Exchange.rate.US..to.NPR.)


# Relationship Mapping Plots


ggplot(Remittance_inflow, aes(x = Remittance, y = Net.migration)) +

  geom_point(color = "blue", size = 3) +

  geom_smooth(method = "lm", se = TRUE, color = "red") +

  labs(title = "Relationship between Remittance and Net Migration", x = "Remittance", y
= "Net Migration") +

  theme_minimal()


data_plot <- Remittance_inflow %>% filter(!is.na(Year), !is.na(Remittance),
!is.na(Net.migration))


ggplot(Remittance_inflow, aes(x = Net.migration, y = GDP)) + geom_point() +
geom_smooth(method = "lm")

ggplot(Remittance_inflow, aes(x = Exchange.rate.US..to.NPR., y = Remittance)) +
geom_point() + geom_smooth(method = "lm")


# Paired Matrices


ggpairs(Remittance_inflow[, c("Remittance", "GDP", "Inflation", "Net.migration",
"Exchange.rate.US..to.NPR.")])


colSums(is.na(Remittance_inflow))


# Normality Check Tests
```

```
shapiro.test(Remittance_inflow$Remittance)
shapiro.test(Remittance_inflow$GDP)
shapiro.test(Remittance_inflow$Inflation)

ggplot(Remittance_inflow, aes(Year, Inflation)) + geom_line() + geom_point()
ggplot(Remittance_inflow, aes(Year, Net.migration)) + geom_line() + geom_point()

# Detailed Corrplot Adjustment

    corrplot(cor_matrix, method = "circle", type = "upper", tl.col = "black",          tl.srt
= 45)

# Checking Outliers

boxplot.stats(Remittance_inflow$Remittance)$out
boxplot.stats(Remittance_inflow$GDP)$out
boxplot.stats(Remittance_inflow$Inflation)$out

# Coefficient of Variation calculations
cv <- data.frame(
  Variable = c("Remittance","GDP","Inflation","NetMigration","ExchangeRate"),
  CV = c(sd(Remittance_inflow$Remittance)/mean(Remittance_inflow$Remittance),
       sd(Remittance_inflow$GDP)/mean(Remittance_inflow$GDP),
       sd(Remittance_inflow$Inflation)/mean(Remittance_inflow$Inflation),
       sd(Remittance_inflow$Net.migration)/mean(Remittance_inflow$Net.migration),
       sd(Remittance_inflow$Exchange.rate.US..to.NPR.)/mean(Remittance_inflow$Exch
ange.rate.US..to.NPR.))
)
```

```
print(cv)

quantile(Remittance_inflow$Remittance)
quantile(Remittance_inflow$GDP)
quantile(Remittance_inflow$Inflation)

write.csv(stats, "EDA_summary.csv")
file.show("EDA_summary.csv")

# Variance Evaluations

var(Remittance_inflow$Remittance)
var(Remittance_inflow$GDP)
var(Remittance_inflow$Inflation)
var(Remittance_inflow$Net.migration)
var(Remittance_inflow$Exchange.rate.US..to.NPR.)
# 1. STATIONARY DATA PREPARATION (The Pro Way)

-
clean_diff_data <- data.frame(
  Year         = Remittance_inflow$Year[-1],
  d_Remittance  = diff(Remittance_inflow$Remittance),
  d_Net_mig     = diff(Remittance_inflow$Net.migration),
  d_Ex_rate     = diff(Remittance_inflow$Exchange.rate.US..to.NPR.),
  g_GDP         = diff(log(Remittance_inflow$GDP)),
  g_Remittance  = diff(log(Remittance_inflow$Remittance)),
  d_Inflation   = diff(Remittance_inflow$Inflation)
)
```

```
clean_diff_data <- na.omit(clean_diff_data)

-
# 2. FIXED ECONOMETRIC REGRESSIONS

model1_fixed <- lm(d_Remittance ~ d_Net_mig + d_Ex_rate, data = clean_diff_data)
model2_fixed <- lm(g_GDP ~ g_Remittance + d_Ex_rate, data = clean_diff_data)
model3_fixed <- lm(d_Inflation ~ d_Remittance, data = clean_diff_data)

d_Inflation  <- c(NA, diff(Remittance_inflow$Inflation))
g_Remittance <- c(NA, diff(log(Remittance_inflow$Remittance)))

model3_data  <- data.frame(d_Inflation, g_Remittance)
model3_data  <- na.omit(model3_data)

model3       <- lm(d_Inflation ~ g_Remittance, data = model3_data)
summary(model3)
dwtest(model3)
bptest(model3)

model4_fixed <- lm(g_GDP ~ g_Remittance, data = clean_diff_data)

stargazer(model1_fixed, model2_fixed, model3_fixed, model4_fixed,
        type = "text", title = "Corrected Regression Results (Stationary Models)")

# 3. PRO DIAGNOSTIC CHECKS

cat("\n--- MULTICOLLINEARITY (VIF < 5 prevents Overfitting) ---\n")
```

```
print(vif(model1_fixed))
print(vif(model2_fixed))

cat("\n--- HETEROSKEDASTICITY (BP Test p > 0.05 is ideal) ---\n")
print(bptest(model1_fixed))
print(bptest(model2_fixed))

cat("\n--- AUTOCORRELATION (DW Test near 2 indicates no tracking errors) ---\n")
print(dwtest(model1_fixed))
print(dwtest(model2_fixed))

cat("\n--- RESIDUAL NORMALITY (JB Test p > 0.05) ---\n")
print(jarque.bera.test(residuals(model1_fixed)))
print(jarque.bera.test(residuals(model2_fixed)))

-


# ALIGNED MACHINE LEARNING SECTION
set.seed(123)
train_index <- createDataPartition(clean_diff_data$g_GDP, p = 0.8, list = FALSE)
train_ml    <- clean_diff_data[train_index, ]
test_ml     <- clean_diff_data[-train_index, ]

tune_grid  <- expand.grid(cp = seq(0.001, 0.1, by = 0.005))
tree_model <- train(g_GDP ~ g_Remittance + d_Ex_rate + d_Net_mig + d_Inflation,
                    data = train_ml, method = "rpart",
                    trControl = trainControl(method = "cv", number = 5),
                    tuneGrid = tune_grid)
```

```
rf_ml_model <- randomForest(g_GDP ~ g_Remittance + d_Ex_rate + d_Net_mig +
d_Inflation,
                            data = train_ml, ntree = 500)


pred_tree <- predict(tree_model, newdata = test_ml)

pred_rf   <- predict(rf_ml_model, newdata = test_ml)


cat("\n--- MACRO MACHINE LEARNING BENCHMARK ---\n")

cat("Decision Tree Test RMSE (GDP Growth):", RMSE(pred_tree, test_ml$g_GDP),
"\n")

cat("Random Forest Test RMSE (GDP Growth):", RMSE(pred_rf, test_ml$g_GDP),
"\n")




# 5. MACRO TIME-SERIES: VAR MODEL


var_ts_data <- ts(clean_diff_data[, c("g_GDP", "d_Remittance", "d_Ex_rate")],
                  start = min(clean_diff_data$Year), frequency = 1)


print(VARselect(var_ts_data))


var_model_fixed <- VAR(var_ts_data, p = 1, type = "const")

summary(var_model_fixed)
```

```
irf_result <- irf(var_model_fixed, impulse = "d_Remittance", response = "g_GDP", n.ahead = 5)

plot(irf_result)


# 6. ARIMA AND FORECASTS EXECUTIONS


remit_ts <- ts(Remittance_inflow$Remittance, start = min(Remittance_inflow$Year), frequency = 1)

print(remit_ts)

plot(remit_ts)


adf.test(Remittance_inflow$Remittance)

diff_remit <- diff(Remittance_inflow$Remittance)

print(diff_remit)

adf.test(diff_remit)


model_arima <- auto.arima(remit_ts)

summary(model_arima)


future <- forecast(model_arima, h = 5)

print(future)

plot(future)


# Granger Causality Testing

grangertest(Remittance ~ Exchange.rate.US..to.NPR., order = 2, data = Remittance_inflow)

grangertest(Remittance ~ Net.migration, order = 2, data = Remittance_inflow)


# Re-specifying Alternative Raw VAR models
```

```
var_data <- Remittance_inflow[, c("Remittance", "GDP", "Inflation")]
var_data <- na.omit(var_data)
colSums(is.na(var_data))
var_data <- ts(var_data, start = min(Remittance_inflow$Year), frequency = 1)

VARselect(var_data)
var_model <- VAR(var_data, p = 1, type = "const")
summary(var_model)
irf_result <- irf(var_model)
print(irf_result$irf)

var_data <- Remittance_inflow[, c("GDP", "Remittance", "Exchange.rate.US..to.NPR.")]
var_data <- na.omit(as.matrix(var_data))
var_model <- VAR(var_data, p = 2, type = "const")
irf_result <- irf(var_model, impulse = "Remittance", response = "GDP", n.ahead = 10)
irf_result

dev.off()
plot(irf_result)

accuracy(model_arima)
checkresiduals(model_arima)

# Secondary Time-Series Model: GDP Auto Arima Tests

gdp_ts <- ts(Remittance_inflow$GDP, start=min(Remittance_inflow$Year),
frequency=1)
gdp_model <- auto.arima(gdp_ts)
sum(is.na(gdp_ts))
```

```
class(gdp_ts)
length(gdp_ts)
summary(gdp_ts)
sum(is.infinite(gdp_ts))
head(gdp_ts)
tail(gdp_ts)
summary(gdp_model)
str(gdp_ts)
summary(gdp_ts)
sum(is.infinite(gdp_ts))
any(gdp_ts <= 0)
plot(gdp_ts)
summary(gdp_ts)
sum(is.infinite(gdp_ts))
any(gdp_ts <= 0)
typeof(gdp_ts)
mode(gdp_ts)
class(gdp_ts)
is.numeric(gdp_ts)

# Re-creating clean numeric vector for GDP Time Series
gdp_ts <- ts(as.numeric(gdp_ts), start = 1993, frequency = 1)
is.numeric(gdp_ts)

gdp_model <- Arima(gdp_ts, order = c(1,1,1))
future_gdp <- forecast(gdp_model, h=5)
plot(future_gdp)
```

```
fevd_result <- fevd(var_model)
plot(fevd_result)


tail(Remittance_inflow[, c("Year", "Remittance", "Net.migration")])


# Standardized Scaling Line Charts Comparisons
plot_data <- Remittance_inflow %>%
  filter(!is.na(Net.migration)) %>%
  mutate(
    Remittance_z = scale(Remittance),
    NetMigration_z = scale(Net.migration)
  )


ggplot(plot_data) +
  geom_line(aes(Year, Remittance_z, colour = "Remittance"), linewidth = 1) +
  geom_line(aes(Year, NetMigration_z, colour = "Net Migration"), linewidth = 1) +
  labs(title = "Standardized Remittance and Net Migration", x = "Year", y =
"Standardized Value (Z-score)") +
  theme_minimal()
```

7. References

- **Remittance Inflow:** Total annual personal remittances received (USD), capturing current cash and in-kind transfers from non-resident to resident households.

1Source: World Bank Data360 (KNOMAD Database)

URL: https://data360.worldbank.org/en/indicator/WB_KNOMAD_MRI

- **Gross Domestic Product (GDP):** Gross Domestic Product in current USD, used to measure national economic output and the relative scale of remittance inflows.

Source: World Bank DataBank (World Development Indicators)

URL: https://databank.worldbank.org/indicator/NY.GDP.MKTP.CD

- **Inflation Rate:** Annual Consumer Price Index (CPI) percentage change, used to track market purchasing power and monitor potential demand-pull inflation from capital inflows.

Source: World Bank Data360 / IMF World Economic Outlook

URL: https://data360.worldbank.org/en/indicator/IMF_WEO_PCPI

- **Exchange Rate:** Official annual USD to NPR foreign exchange reference rate, used to analyze how currency valuation affects the real value of incoming transfers.

Source: Nepal Rastra Bank (NRB) Forex Portal

URL: https://www.nrb.org.np/forex

- **Net Migration:** The annual net balance of international migrants (immigrants minus emigrants), serving as a demographic proxy for labor export dynamics.

Source: World Bank Open Data (UN Population Division Estimates)

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